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Thermal-mechanical induced joining of SiO_2/Brass by UV nanosecond laser: modeling and experimental investigation

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Highlights

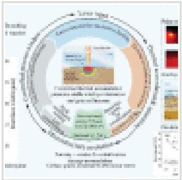
Outline

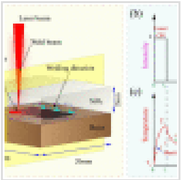
- SiO_2 and brass are directly joined by UV nanosecond laser transmission welding.
- Numerical results showed interfacial peak temperatures reached up to ~ 3200 K.
- Preferential Zn oxidation promotes oxide-assisted wetting and Cu–O–Si chemical bonding.
- Thermal accumulation and saturation are key parameters governing interfacial stability and joint performance.

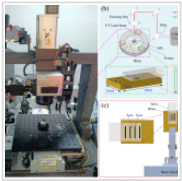
Abstract

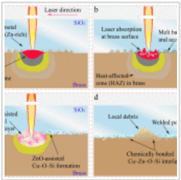
Ultraviolet (UV) nanosecond laser transmission welding is employed to join SiO_2 and brass directly without filling interlayers or external fixtures, and the governing thermal–induced microstructural mechanisms are systematically quantified through combined experiments and three-dimensional transient simulations. Using a 355 nm laser operating at 150 kHz, strong pulse overlap (>95%) induced pronounced thermal accumulation at the glass–metal interface. Numerical results showed that interfacial peak temperatures increased from approximately 2200 K at 1.2 W to 2600 K at 1.6 W and ~ 3200 K at 2.0 W, accompanied by a transition from transient heating to conduction-controlled thermal saturation. Under moderate thermal accumulation (1.2 W, 20 mm·s^{-1}, four passes), localized brass melting occurred without bulk damage to SiO_2, enabling effective micro-gap collapse and formation of a continuous fusion zone. SEM, EDS, and XPS analyses reveal a thin, chemically graded Cu–Zn–O–Si interlayer, where preferential Zn oxidation promotes oxide-assisted wetting and Cu–O–Si chemical bonding. EBSD characterization showed significant grain refinement in the interfacial brass, with an average grain size of ~ 3.3 μm, low kernel average misorientation (KAM < 5°), and a high fraction of high-angle grain boundaries, indicating effective strain accommodation. Mechanical testing demonstrated a non-monotonic dependence of joint strength on laser power, with a maximum shear strength of ~ 28 MPa at 1.2 W, decreasing to ~ 22 MPa at 1.6 W and ~ 14 MPa

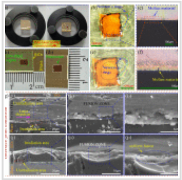
Figures (12)

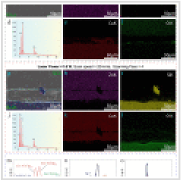






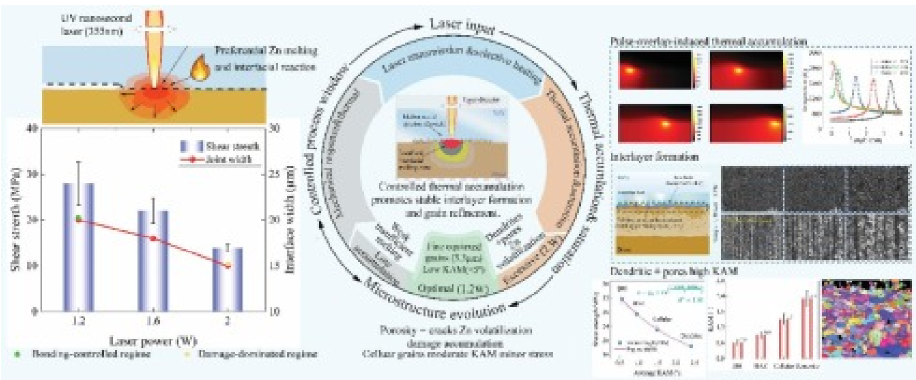






at 2.0W. Excessive thermal accumulation led to Zn volatilization, porosity formation, elevated KAM, and damage-dominated fracture. The experimental trends accurately matched with a coupled thermal–mechanical model linking bonding width, time above glass softening temperature, and residual stress-induced damage. These results established thermal accumulation and saturation as the key parameters governing interfacial stability and joint performance in UV nanosecond laser welding of SiO_2/brass.

Graphical abstract



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Keywords

UV nanosecond laser transmission welding; Glass–metal joining; SiO_2/brass interface; Oxide-assisted wetting; COMSOL simulation

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Nomenclature

Symbol
: Description, Units

T
: Temperature, K

T_a
: Ambient temperature, K

T_i
: Initial temperature, K

T_s
: Glass softening temperature, K

ρ
: Density, g·cm^{-3}

c_p
: Specific heat capacity, J·g^{-1}·K^{-1}

k
: Thermal conductivity, W·m^{-1}·K^{-1}

Q
: Laser heat input, W

L_m
: Latent heat of melting, J·g^{-1}

α
: Thermal expansion coefficient, K^{-1}

μ
: Dynamic viscosity, Pa·s

v
: Laser scanning speed, mm·s^{-1}

P
: Average laser power, W

E_p
: Pulse energy, μJ

f
: Laser repetition rate, Hz

D
: Laser beam diameter, μm

A
: Laser absorptivity, –

ε
: Surface emissivity, –

h
: Heat transfer coefficient, W·m^{-2}·K^{-1}

x_r
: Laser beam center position, μm

E_d
: Average laser energy density, J·m^{-2}

ϕ

Gaussian beam standard deviation, μm

ΔT
: Half-width of temperature curve, K

n
: Number of scanning passes, –

σ_r
: Residual thermal stress MPa

σ_f
: Fracture strength of glass, MPa

τ
: Joint shear strength, MPa

τ_{sat}
: Saturation shear strength, MPa

w
: Effective bonding width, μm

t_b
: Time above glass softening temperature, s

Abbreviations

UV
: Ultraviolet

SEM
: Scanning electron microscopy

EDS

Energy-dispersive X-ray spectroscopy

XPS
: X-ray photoelectron spectroscopy

EBSD
: Electron backscatter diffraction

COMSOL
: COMSOL Multiphysics®

HAZ
: Heat-affected zone

FZ
: Fusion zone

BM
: Base material

LAIZ
: Laser-affected interfacial zone

LIML
: Laser-induced metallurgical mixing layer

IFZ
: Interfacial fracture zone

KAM
: Kernel average misorientation

NDI
: Normalized damage index

FCC
: Face-centered cubic

BCC
: Body-centered cubic

1. Introduction

Transparent brittle materials such as silicon-dioxide (SiO_2, glass) and optical crystals play a vital role in modern engineering systems, including biomedical devices, aerospace structures, consumer electronics, and microwave technologies, due to their excellent optical transparency, high mechanical stiffness, superior corrosion resistance, and outstanding thermal stability [1], [2]. In many of these applications, SiO_2 components must be integrated with metallic materials to enable electricaltransfer, thermal management, and mechanical reinforcement. Brass is particularly attractive for this purpose owing to its favorable combination of electrical and thermal conductivity, corrosion resistance, machinability, and ductility [3]. Consequently, the development of reliable SiO_2-brass joints is of increasing importance for electronic packaging, optoelectronic assemblies, and microelectromechanical systems (MEMS) [4]. The primary difficulty in joining SiO_2 to brass stems from the pronounced mismatch in their physical and chemical properties, including melting temperature, thermal conductivity, coefficient of thermal expansion, and optical absorption behavior. Conventional joining techniques such as cementation [5], conventional joining techniques such mechanical fastening [6], diffusion bonding [7], and brazing, often suffer from inherent limitations, including weak interfacial bonding, poor spatial precision, extensive thermal damage, and limited long-term reliability. In glass-metal systems, thick interlayers and prolonged thermal exposure can introduce excessive residual stresses and interfacial defects, significantly degrading joint strength and long-term reliability [8]. These limitations have motivated the development of advanced joining technologies capable of producing high-strength, precise, and defect-free SiO_2-metal joints.

Laser-based joining has emerged as a promising alternative due to its non-contact nature, high energy density, and excellent spatial and temporal controllability. By exploiting nonlinear absorption and localized energy deposition at the glass-metal interface, pulsed laser welding enables effective joining of transparent brittle materials with reduced heat-affected zones compared with conventional thermal processes [9]. However, the formation of brittle reaction layers and undesirable intermetallic compounds remains a concern in dissimilar material welding under rapid thermal cycling [10], [11]. Ultrashort pulse lasers, including femtosecond (fs) and picosecond (ps) systems, have attracted considerable attention for dissimilar material welding due to their ability to confine energy deposition within extremely short timescales, thereby minimizing heat diffusion and collateral thermal damage [12]. Numerous studies have demonstrated the feasibility of fs-laser welding for metal–glass systems. For instance, Zhang et al. [13] systematically investigated the effects of processing energy and scanning speed on the shear strength of femtosecond laser-welded Cu/SiO_2 joints, achieving a maximum bonding strength of 2.34MPa. Wang et al. [14] reported a shear strength of 8.79MPa for stainless steel-glass joints, while Li et al. [15] achieved a peak strength of 10.9MPa for copper-glass bonding. Despite these encouraging results, the high system cost, limited processing efficiency, and scalability challenges associated with ultrashort pulse lasers restrict their widespread industrial adoption [16], [17]. As a more cost-effective and industrially scalable alternative, nanosecond-pulsed lasers have received increasing attention while retaining the core advantages of laser transmission welding. Recent studies have demonstrated that strong glass-metal joints can be achieved using nanosecond lasers when processing parameters are carefully optimized. Huo et al. [18], reported a shear strength of up to 16.39MPa for silica glass-304 stainless steel joints fabricated using an infrared nanosecond fiber laser. These results demonstrate that nanosecond laser welding enables controlled interfacial reactions, localized melting, and mechanical interlocking, thereby enhancing joint strength and reliability in glass-metal systems [11].

Accurate thermal modeling is essential for understanding and optimizing laser welding processes, as the transient temperature field governs melt pool evolution, thermal stress development, phase transformations, and interfacial reactions. Finite-element simulations, such as those performed using COMSOL Multiphysics, require reliable prediction of temperature distributions to incorporate phase change, temperature-dependent material properties, and reaction kinetics at the joint interface. Classical analytical models, including Rosenthal's moving heat source solution [19], [20], provide a foundational framework for welding simulations. Hartwig et al. [21] proposed a physically motivated formulation based on melt pool dynamics and absorption behavior rather than empirical geometric assumptions. In addition, Gaussian volumetric heat source models incorporating phase-change effects have been successfully validated against experimental melt pool geometries [22], with further refinements reported to improve predictive accuracy for pulsed laser welding processes [23], [24]. However, these conventional models often fail to capture the highly localized, transient, and non-equilibrium energy deposition associated with UV nanosecond laser irradiation in dissimilar material systems.

Despite recent progress, the fundamental mechanisms governing interfacial stability and strength under high-repetition-rate nanosecond laser irradiation remain incompletely understood. In particular, the roles of pulse-overlap-induced thermal accumulation, thermal saturation, and their coupling with interfacial reactions, microstructural evolution, and mechanical failure have not been systematically clarified for SiO_2/brass interfaces. Most existing studies focus on empirical correlations between process parameters and joint strength, without establishing a unified thermal-microstructural framework capable of defining a stable processing window. In this work, UV nanosecond laser transmission welding of SiO_2 and brass is investigated through a combined experimental and numerical approach. A 3D transient heat-transfer model is developed to quantify interfacial thermal accumulation and saturation under multi-pulse irradiation. The resulting interfacial morphology, chemical bonding, and microstructural evolution are systematically characterized using SEM, EDS, XPS, and EBSD, and correlated with joint mechanical performance. By linking thermal accumulation to the transition from strain-tolerant, oxide-assisted interfacial bonding to damage-dominated failure, this study establishes a mechanistic framework for defining an optimal processing window in nanosecond laser glass-metal welding.

2. Theoretical modeling

A 3D transient thermal–mechanical model is developed to describe laser-induced heating and joining at the brass/SiO_2 interface under multi-pulse irradiation. The model is formulated to quantify thermal accumulation, interfacial softening, residual stress development, and their combined influence on joint shear strength. Rather than resolving highly localized, non-equilibrium optical hotspots, the formulation focuses on the bulk-averaged joining temperature, which governs glass softening, viscous flow, and bond formation. The modelling framework combines a moving laser heat source, cumulative thermal effects from multiple scan passes, and asymmetric heat diffusion into glass and brass within a finite thermally affected zone (TAZ). Based on the predicted thermal history, a shear strength model is subsequently developed that accounts for both bonding enhancement and crack-induced degradation. This reduced-order approach provides a physically meaningful and computationally efficient description of laser glass–metal joining.

2.1. Assumptions

The following assumptions are adopted in the development of the thermal and shear-strength models:

a) Heat transfer within both glass (SiO_2) and brass are conduction dominated; convective and radiative heat losses are neglected due to the short interaction time and localized heating.

b) The laser is modeled as a moving surface heat source with constant scan speed and constant average power; acceleration effects at the scan boundaries are neglected.

c) The joining temperature is defined as a bulk-averaged temperature within the glass, rather than the instantaneous optical hotspot at the interface.

d) Perfect thermal contact is assumed at the glass-brass interface, and temperature is continuous across the interface.

e) The thermal effect of multiple scan passes is represented by linear superposition, enabling modeling of thermal accumulation and saturation.

f) Glass (SiO_2) and brass are assumed isotropic and homogeneous, with thermophysical properties taken as constant over the joining temperature range.

g) The glass layer is assumed optically transparent at the laser wavelength, and laser energy is primarily absorbed at the brass surface and conducted into the glass.

h) Joint failure is assumed to be glass-limited, and the influence of fluid flow, Marangoni convection,

The laser spot diameter $D = 3\mu m$ represents the optical energy-deposition scale, whereas $R_{\mathrm{TAZ}} = 100\,\mu m$ is defined as an effective conduction-controlled thermal averaging radius. Under high-repetition-rate irradiation (150kHz, >95% overlap), thermal accumulation leads to a diffusion length ‘$L \approx \sqrt{\alpha\tau}$’ that exceeds the optical spot, so this spatial averaging reduces the instantaneous peak hotspot temperature but more realistically captures the bulk interfacial joining temperature over the finite thermally affected zone, consistent with established conduction-based scaling and distributed heat-source modeling approaches [25], [26].

2.2. Thermal model

The transient temperature field in each solid domain is governed by the three-dimensional heat conduction equation [26];

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q(x, y, t) \tag{1}$$

where (T) is the temperature, (\rho) the density, (c_p) the specific heat capacity, (k) the thermal conductivity, and (Q) the laser heat input. The computational domain, laser scanning direction, and applied boundary conditions used in the numerical model are illustrated in Fig. 1(a). The figure highlights the glass–brass configuration, the heat-affected zone, and the dominant heat-transfer paths within the assembly.

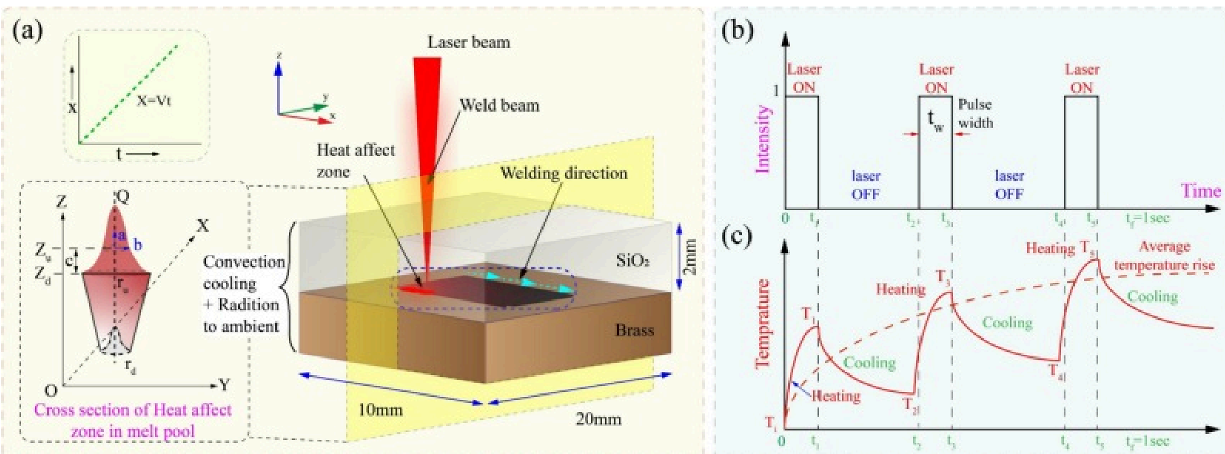

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Fig. 1. Schematic illustration of the laser welding configuration and thermal modeling framework. (a) 3D schematic of the glass-brass laser welding setup, indicating the laser beam, welding direction, heat-affected zone, and boundary conditions used in the numerical model, together with the representation of the conical surface heat flux and heat transfer by conduction, convection, and radiation. (b) Temporal profile of pulsed laser irradiation, showing laser on/off cycles and pulse width. (c) Corresponding transient temperature evolution during pulsed laser heating, illustrating repeated heating and cooling cycles and progressive thermal accumulation under successive laser pulses.

The laser scans along the (x)-direction with constant velocity (v), such that the instantaneous beam center is given by [27];

$$x_{\mathrm{L}}(t)= vt \tag{2}$$

For a UV laser wavelength of 355 nm, the optical penetration depth in brass is on the order of tens of nanometers, which is several orders of magnitude smaller than the numerical mesh size. Consequently, laser energy absorption is effectively confined to the brass surface. The laser heat input is therefore modeled as a Gaussian surface heat flux, corresponding to the asymptotic limit of a volumetric Beer-Lambert absorption model for strong absorption. The applied surface heat flux is expressed as [28];

$$q_{\mathrm{laser}}(x,y,t)= I_0(t)exp!\left(-\frac{2r(x,y,t)^2}{\omega^2}\right) \tag{3}$$

where,

$$r(x,y,t)= \sqrt{\left(x-x_{\mathrm{L}}(t)\right)^2+\left(y-y_0\right)^2} \tag{4}$$

and $(\omega = D/2)$ is the beam radius corresponding to the $(1/e^2)$ intensity level. For thermal accumulation analysis, a time-averaged intensity formulation is employed,

$$I_{0,\mathrm{avg}} = \frac{4P_{\mathrm{avg}}}{\pi D^2} \tag{5}$$

which captures the cumulative thermal effect of high-repetition-rate pulsed irradiation without explicitly resolving individual nanosecond pulses [29]. The pulsed nature of the laser irradiation and the resulting transient temperature evolution at the interface are schematically shown in Fig. 1(b). Each laser pulse produces rapid heating followed by cooling during the pulse-off period, and incomplete cooling between successive pulses leads to progressive thermal accumulation. A conceptual comparison between a volumetric conical heat-source formulation based on Beer–Lambert absorption and its surface heat-flux limit is shown in Fig. 1(c). Owing to the extremely small optical penetration depth of 355 nm radiation in brass, laser energy deposition is effectively confined to the surface, and the surface heat-flux formulation is adopted in the present model.

To avoid singular behavior associated with point heat-source formulations, the joining temperature is defined as a bulk-averaged temperature over a finite thermally affected zone characterized by a lateral radius (R_{TAZ}). The peak bulk temperature rise at the interface due to a single scan pass is given by;

$$\Delta T_{\mathrm{peak}} = \frac{\eta P}{2\pi k_g R_{\mathrm{TAZ}}} \tag{6}$$

where 'η', is the effective absorptivity and 'k_g', is the thermal conductivity of glass, which governs the joining temperature due to its lower thermal diffusivity. For multiple scan passes 'N_p', thermal accumulation is modeled as;

$$\Delta T_{\mathrm{peak}}^{(N_p)} = N_p, \Delta T_{\mathrm{peak}} \tag{7}$$

The corresponding bulk interfacial temperature is:

$$T_{\mathrm{interface}} = T_0 + \Delta T_{\mathrm{peak}}^{(N_p)} \tag{8}$$

where (T_0) is the ambient temperature.

The spatial distribution of the bulk joining temperature along the scan direction is described by a Gaussian profile centered at the instantaneous laser position [30], [31],

$$T(x,t) = T_0 + \Delta T_{\mathrm{peak}}^{(N_p)} exp! \left[- \frac{(x - x_{\mathrm{L}}(t))^2}{2R_{\mathrm{TAZ}}^2} \right] \tag{9}$$

which captures the motion of the thermal peak with the scanning laser and the finite width of the joining zone. Heat diffusion normal to the interface is modeled using exponential decay functions, reflecting diffusion-controlled thermal penetration.

For the glass side ($z \geq 0$):

$$T_g(x,z,t) = T_0 + [T(x,t) - T_0] exp\left(-\frac{z}{Z_g}\right) \tag{10}$$

For the brass side ((z<0)):

$$T_b(x,z,t) = T_0 + [T(x,t) - T_0] exp\left(\frac{z}{Z_b}\right) \tag{11}$$

where (Z_g) and (Z_b) are characteristic thermal penetration depths in glass and brass, respectively. Combining Eqs. (9)-(11), the three-dimensional transient bulk temperature field is obtained as;

$$T(x,z,t) = T_0 + \Delta T_{\text{peak}}^{(N_p)} \exp\left[-\frac{(x - x_{\mathrm{L}}(t))^2}{2R_{\mathrm{TAZ}}^2}\right] \times \begin{cases} \exp\left(-\frac{z}{Z_g}\right), z \geq 0 \\ \exp\left(-\frac{z}{Z_g}\right), z < 0 \end{cases} \tag{12}$$

This expression forms the basis for the two-dimensional and three-dimensional temperature fields analyzed in this study. Laser joining is considered feasible when the bulk temperature in the glass exceeds the softening temperature,

$$T(x,z,t) \geq T_{\text{soft}} \tag{13}$$

which defines the joining window in terms of laser power, scan speed, scan length, and number of scan passes.

2.3. Shear strength model

While the thermal model quantifies the extent of glass softening, experimental lap-shear tests reveal that joint shear strength exhibits a non-monotonic dependence on laser energy input [32]. To unify the effects of laser power, scan speed, and number of scan passes, an effective laser energy density is defined as;

$$E_d = \frac{N_p P}{v} \tag{14}$$

From the thermal field, two bonding metrics are extracted: the effective bonding width (W_b), defined by Eq. (13), and the time above softening temperature (τ_{soft}). The bonding-controlled shear strength is modeled as;

$$\tau_{\text{bond}} = \tau_{max}\left[1 - exp!\left(-\frac{\tau_{\text{soft}}}{\tau_0}\right)\right]\frac{W_b}{W_0} \tag{15}$$

where τ_{max} is the saturation shear strength, (τ_0) is a characteristic bonding time constant, and (W_0) is a reference bonding width.

At high energy densities, residual thermal stress develops in the glass due to thermal expansion mismatch during cooling [33], [34]. The residual stress is approximated as;

$$\sigma_{\text{res}} = E_g\alpha_g\Delta T_{\text{peak}}^{(N_p)} \tag{16}$$

Here, Eq. (16) is used as a simplified effective approximation of residual thermal stress. The true stress field at the SiO_2/brass interface is 3D, because of constrained thermal expansion and non-uniform cooling [7]. Thus, this term is treated as a phenomenological stress indicator in the present damage model.

Crack initiation is assumed to occur when this residual stress exceeds the fracture strength of the glass, defining a critical energy density ($E_{d,c}$). Strength degradation due to cracking is represented by a damage factor,

$$D(E_d) = \exp[-\beta\max(0, E_d - E_{d,c})] \tag{17}$$

In the high-energy regime, Zn-volatilization-induced instability and associated porosity are not explicitly simulated, instead, their combined effect is incorporated phenomenologically through the damage term [15].

Experimental results further indicate that joint strength exceeds predictions based solely on thermal softening and residual stress. This enhancement is attributed to additional interfacial mechanisms such as chemical bonding and mechanical interlocking and is incorporated through an interfacial enhancement factor,

$$F_{\text{int}} = 1 + \gamma \frac{\tau_{\text{soft}}}{\tau_{\text{soft,max}}} \tag{18}$$

The final joint shear strength is therefore expressed as;

$$\tau(E_d) = \tau_{max}\left[1 - exp!\left(-\frac{\tau_{\text{soft}}(E_d)}{\tau_0}\right)\right]\frac{W_b(E_d)}{W_0}exp[-\beta \max(0, E_d - E_{d,c})](1 + \gamma \frac{\tau_{\text{soft}}}{\tau_{\text{soft,max}}}) \tag{19}$$

In the present model, τ_0, β, and γ are treated as phenomenological fitting parameters and were calibrated against the experimental lap-shear results, whereas E_g, σ_c, and α_{eff} were assigned based on material properties and physical considerations.

This formulation predicts the experimentally observed non-monotonic dependence of shear strength on laser energy density. At low energy density, insufficient glass softening results in negligible bonding and low shear strength. At intermediate energy density, optimal softening and wetting produce maximum joint strength. At high energy density, residual thermal stress and crack formation reduce effective load transfer, leading to strength degradation. The saturation shear strength τ_{max} is governed by the intrinsic shear strength of the glass, typically in the range of 40–80 MPa for fused silica and high-purity SiO_2, consistent with experimental fracture observations.

3. Experimental

3.1. Materials and sample preparation

Brass copper alloy CuZn39Pb3 specimens and double-side-polished SiO_2 substrates were used as the base materials. The SiO_2 substrates had dimensions of 20×20×2 mm, while the brass sheets were cut to matching sizes to ensure full overlap during welding. Prior to joining, the brass samples were mechanically polished using progressively finer sandpapers to remove surface oxides and contaminants. The polished samples were then ultrasonically cleaned in ethanol for 25 min, rinsed thoroughly with distilled water, and dried in air to obtain clean, oxide-free surfaces. The SiO_2 substrates were cleaned using a standard solvent cleaning. Selected thermophysical and mechanical properties of the experimental materials and model parameters are summarized in Table 1 and Table 2, including melting and boiling temperatures, thermal conductivity, specific heat capacity, thermal expansion coefficient, density, and refractive index. These parameters were used for both experimental analyses and numerical simulations.

Table 1. Thermophysical and mechanical properties of SiO_2 and brass.

Property	Symbol	SiO_2 (Fused silica)	Brass (CuZn39Pb3)	Units
Melting temperature	T_m	1715	875–890	°C
Boiling temperature	T_b	2230	~1800	°C
Density	ρ	2.65	8.96	$g{\cdot}cm^{-3}$
Thermal conductivity	k	1.4	401	$W{\cdot}m^{-1}{\cdot}K^{-1}$
Specific heat capacity	C_p	0.703	0.385	$J{\cdot}g^{-1}{\cdot}K^{-1}$
Latent heat of melting	L_m	1800	205	$J{\cdot}g^{-1}$
Thermal expansion coefficient	α	9×10^{-6}	$19–21\times10^{-6}$	K^{-1}
Young's modulus	E	~70	~100	GPa
Refractive index	n	1.458	–	–
Dynamic viscosity*	μ	~0.1 (softened)	–	Pa·s

Table 2. Mechanical and shear-strength model parameters.

Parameter	Symbol	Value	Units	Description
Young's modulus of glass	E_g	70	GPa	Elastic modulus of SiO_2

Parameter	Symbol	Value	Units	Description
Fracture strength of glass	σ_c	~40	MPa	Tensile fracture strength
Effective thermal expansion parameter	α_{eff}	9×10^{-6}	K^{-1}	Constrained interfacial thermal strain
Saturation shear strength	$\tau_m ax$	50	MPa	Upper bonding limit
Characteristic bonding time	τ^0	0.004	s	Bond formation timescale
Damage coefficient	β	0.008	$(J\cdot m^{-1})^{-1}$	Energy-based degradation
Interfacial enhancement factor	γ	1	–	Chemical / mechanical bonding

3.2. Experimental setup and joining procedure

Laser joining was performed using a Coherent AVIA LX 355 nm ultraviolet nanosecond laser system, as shown in Fig. 2(a). The laser operated with a pulse width of 20–50 ns, a maximum pulse energy of 30 μJ, and a repetition rate ranging from 1 to 200 kHz. During joining process, the SiO_2 substrate was placed on top of the brass sheet in an overlap configuration. A custom-designed fixture incorporating two convex lenses was employed to suppress optical interference effects (Newton rings) and to maintain a controlled and uniform interface gap between the SiO_2 and brass surfaces. A schematic illustration of the optical path, specimen configuration, and scanning strategy is shown in Fig. 2(b).

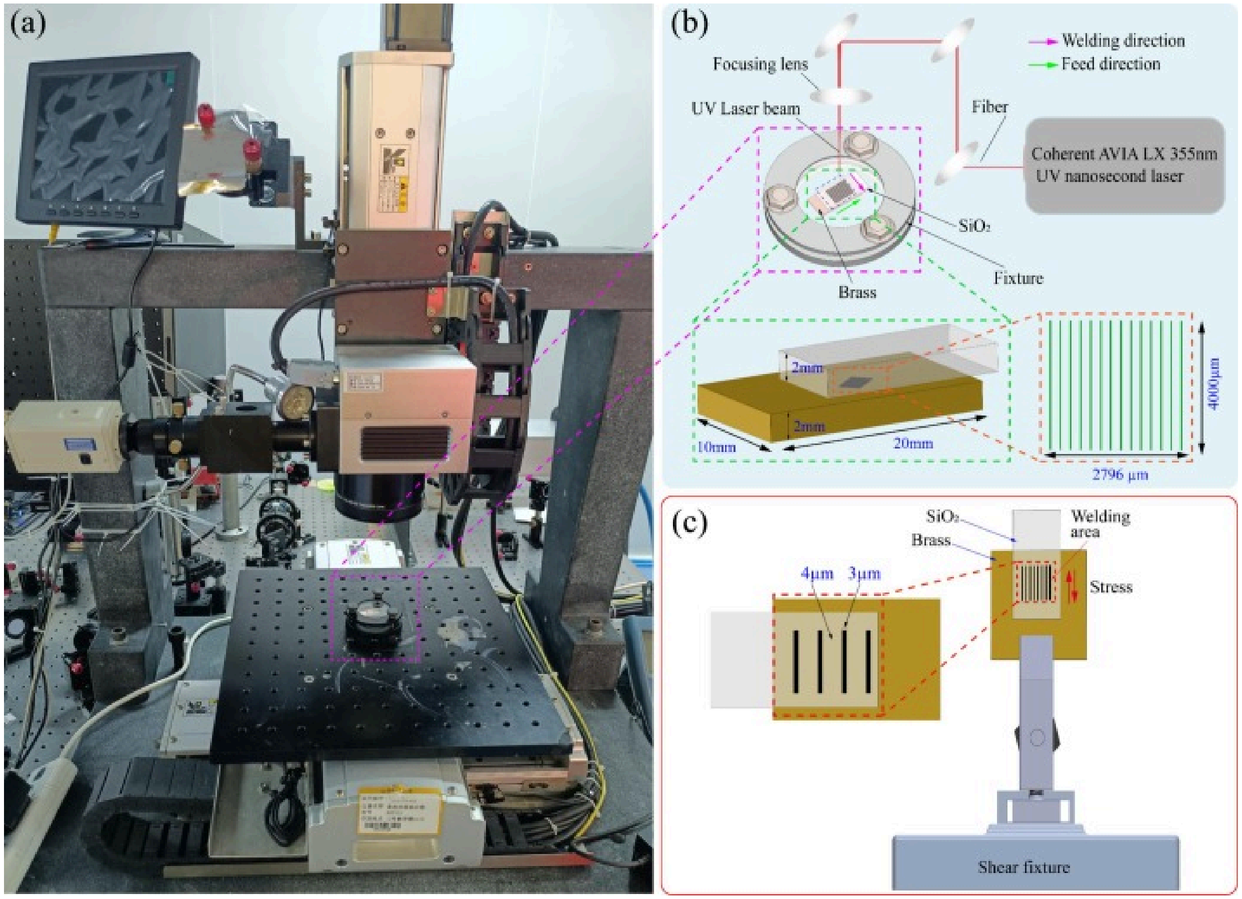

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Fig. 2. Experimental setup and schematic illustration mechanical testing. (a) Photograph of the AVIA LX 355 nm nanosecond UV laser welding system. (b) Schematic of the laser–material interaction and scanning strategy, showing the optical path, specimen configuration, welding direction, and hatch scanning pattern. (c) Schematic illustration of the shear strength testing configuration, indicating the welded area, loading direction.

The laser beam was transmitted through the transparent SiO_2 substrate and absorbed at the brass surface, resulting in localized heating, interfacial melting, and bonding. The laser was scanned over a rectangular area of $(3\,\mu m+4\,\mu m)\times 4000\,\mu m$ using a hatch scanning strategy; where $3\,\mu m$ is only one laser beam width without scanning with a line spacing of $4\,\mu m$. Laser power levels of 1.2, 1.6, and 2 W were combined with scanning speeds of 20, 18, and $15\,mm\cdot s^{-1}$. Each weld area was scanned 2, 3, or 4 times to investigate the effect of cumulative thermal input. Due to the pulsed nature of the nanosecond laser, cyclic thermal accumulation occurred during processing, as successive laser pulses were delivered before the interface fully cooled. This effect contributed to a gradual temperature rise and promoted stable bond formation at the SiO_2/brass interface. The laser processing parameters and thermal boundary conditions used in both the experiments and numerical simulations are summarized in Table 3.

Table 3. Laser processing and thermal modeling parameters.

Parameter	**Symbol**	**Value**	**Units**
Laser wavelength	λ	355	nm
Pulse repetition rate	f	150 kHz	Hz
Pulse width	τ_p	20–50	ns
Maximum pulse energy	E_p	30	μJ
Average laser power	P	1.2, 1.6, 2.0	W
Scanning speed (experiment)	v	20, 18, 15	$mm\cdot s^{-1}$
Scanning speed (thermal model)	v_0	10	$mm\cdot s^{-1}$
Number of scanning passes	N_p	2–4 (exp.), 3 (model)	–
Beam diameter	D	3	μm
Absorptivity	η	0.25–0.7	–
Ambient temperature	T_0	300	K
Glass softening temperature	T_{soft}	1988	K

Parameter	Symbol	Value	Units
Bulk averaging radius	R_{TAZ}	100	μm
Effective glass penetration depth	Z_g	30	μm
Effective brass penetration depth	Z_b	200	μm

3.3. Mechanical testing and microstructural characterization

The shear strength of the welded joints was evaluated using a high-precision micro-force testing system. Tests were conducted at a constant displacement rate of 0.4mm·min^{-1}. A schematic of the specimen geometry and shear testing fixture is shown in Fig. 2(c), illustrating the welded area, loading direction, and stress distribution. For each set of laser processing conditions, three specimens were tested to ensure repeatability, and the average shear strength was calculated. Following mechanical testing, fracture surfaces were examined using scanning electron microscopy (SEM) to identify failure modes and interfacial features. After separation of the welded joints, X-ray photoelectron spectroscopy (XPS) was employed to analyze the chemical bonding states and elemental composition at the fracture surfaces. For metallographic and crystallographic analysis, cross-sectional brass specimens were mechanically polished and etched for approximately 30s using a solution containing 4mL $FeCl_3$, 5mL HCl, and 30mL distilled water to reveal microstructural features. Subsequent fine polishing was performed using 800, 1200, 1500, 2000, and 3000 grit sandpapers.

Electropolishing was then carried out using a 10% perchloric acid–ethanol ($HClO_4$-C_2H_5OH) electrolyte at −25°C, maintained with liquid nitrogen, under an applied voltage of 20V. After electropolishing, the samples were examined by SEM and electron backscatter diffraction (EBSD). During EBSD measurements, the specimens were tilted by 70° to enhance backscattered electron signal quality and obtain reliable crystallographic orientation data.

4. Results and discussion

4.1. Working principle

Fig. 3 illustrates the dominant physical mechanisms governing UV nanosecond laser transmission welding of SiO_2 to brass. During processing, a 355 nm nanosecond laser pulse train is transmitted through the transparent SiO_2 and selectively absorbed at the brass surface, where rapid photothermal conversion generates highly localized interfacial heating and melting. Owing to the significantly lower melting point of Zn (~420 °C) compared with Cu (~1085 °C), Zn preferentially enters the liquid phase, resulting in the formation of a Zn-rich molten layer at the interface Fig. 3(a). Similar preferential melting and redistribution of Zn in laser-processed Zn-containing alloys have been widely reported [11].Under the applied low-power nanosecond irradiation conditions, recoil pressure induced by vaporization remains weak and does not dominate melt pool behavior. Instead, surface-tension gradients caused by temperature and compositional differences drive Marangoni convection [35], leading to lateral melt redistribution and limited molten metal displacement along the interface Fig. 3(a).

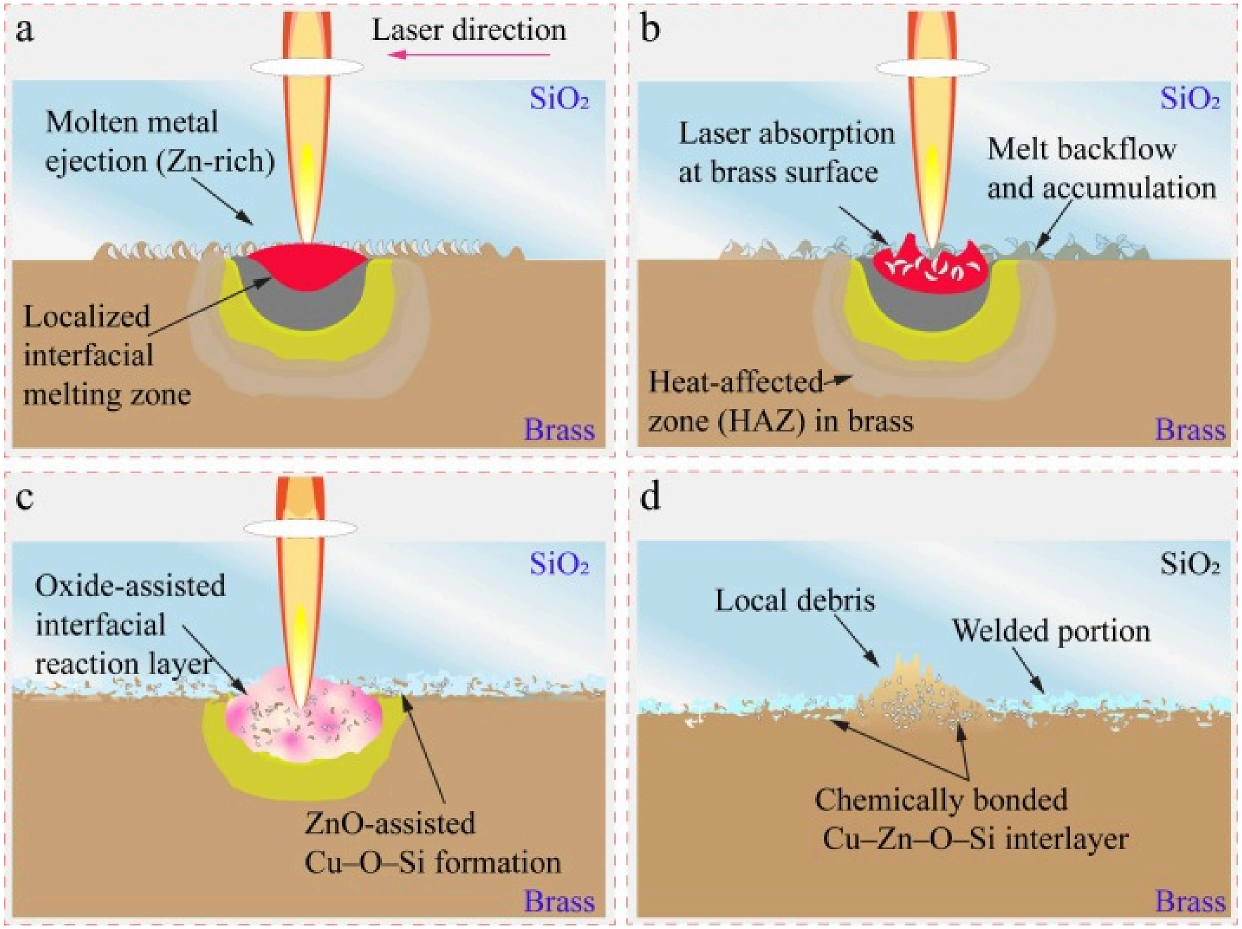

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Fig. 3. Schematic illustration of the joint formation mechanism during UV nanosecond laser transmission welding of SiO_2/brass. (a) Laser transmission through SiO_2 and localized absorption at the brass surface, inducing Zn-rich interfacial melting and molten metal. (b) Pulse-overlap-induced thermal accumulation leading to melt backflow and formation of a heat-affected zone (HAZ). (c) Oxide-assisted interfacial reactions promoted by preferential Zn oxidation, resulting in ZnO-assisted Cu–O–Si chemical bonding. (d) A continuous, chemically bonded Cu–Zn–O–Si interlayer local debris residues.

Due to the high pulse overlap associated with the 150 kHz repetition rate and low scanning speed, pronounced thermal accumulation occurs. This cumulative heating enhances melt backflow toward previously irradiated regions and promotes the formation of a localized heat-affected zone (HAZ) within the brass substrate Fig. 3(b). Comparable pulse-overlap-induced thermal accumulation and intensified melt dynamics have been observed in high-repetition-rate nanosecond laser welding studies [36]. While the interface remains molten, ambient oxygen participates in oxide-assisted interfacial reactions. Preferential oxidation of Zn improves interfacial wetting and chemical activity, enabling the formation of ZnO and Cu–O–Si bonding species at the metal–glass interface Fig. 3(c), consistent with oxide-mediated bonding mechanisms reported in laser joining of metals and silicate glasses. Upon cooling after the final overlapping pass, the joint solidifies into a spindle-shaped weld seam containing a continuous, chemically bonded Cu-Zn-O-Si interlayer Fig. 3(d), which enhances load transfer and joint strength, as demonstrated in previous metal–glass laser welding studies [37].

4.2. Macroscopic and microstructure analyses of UV laser welded SiO_2/brass interfaces

Fig. 4 presents the macroscopic morphology of SiO_2/brass joints fabricated by UV nanosecond laser transmission welding, where a well-defined square fusion region corresponding to the laser scanning path is obtained without the use of external fixtures [11]. No macroscopic defects such as cracking, delamination, or spallation are observed within the welded region in Fig. 4(a-c), indicating that the selected laser parameters provide stable bonding while limiting excessive thermal stress in the brittle SiO_2 substrate [34]. Prior to laser irradiation, distinct Newton's rings were observed at the SiO_2/brass interface as seen in Fig. 4(d_1) and Fig. 4(e), confirming the presence of a nanometrically interfacial air gap and non-optical contact between the two surfaces [38]. The observation of Newton's rings prior to irradiation further examined the initial separation between the brass and SiO_2 surface's optical-contact threshold (typically $<\lambda/4$). These interference fringes originate from multiple-beam interference within a wedgeshaped thin film and are commonly used to characterize initial gap distributions in glass-metal contact systems [39].

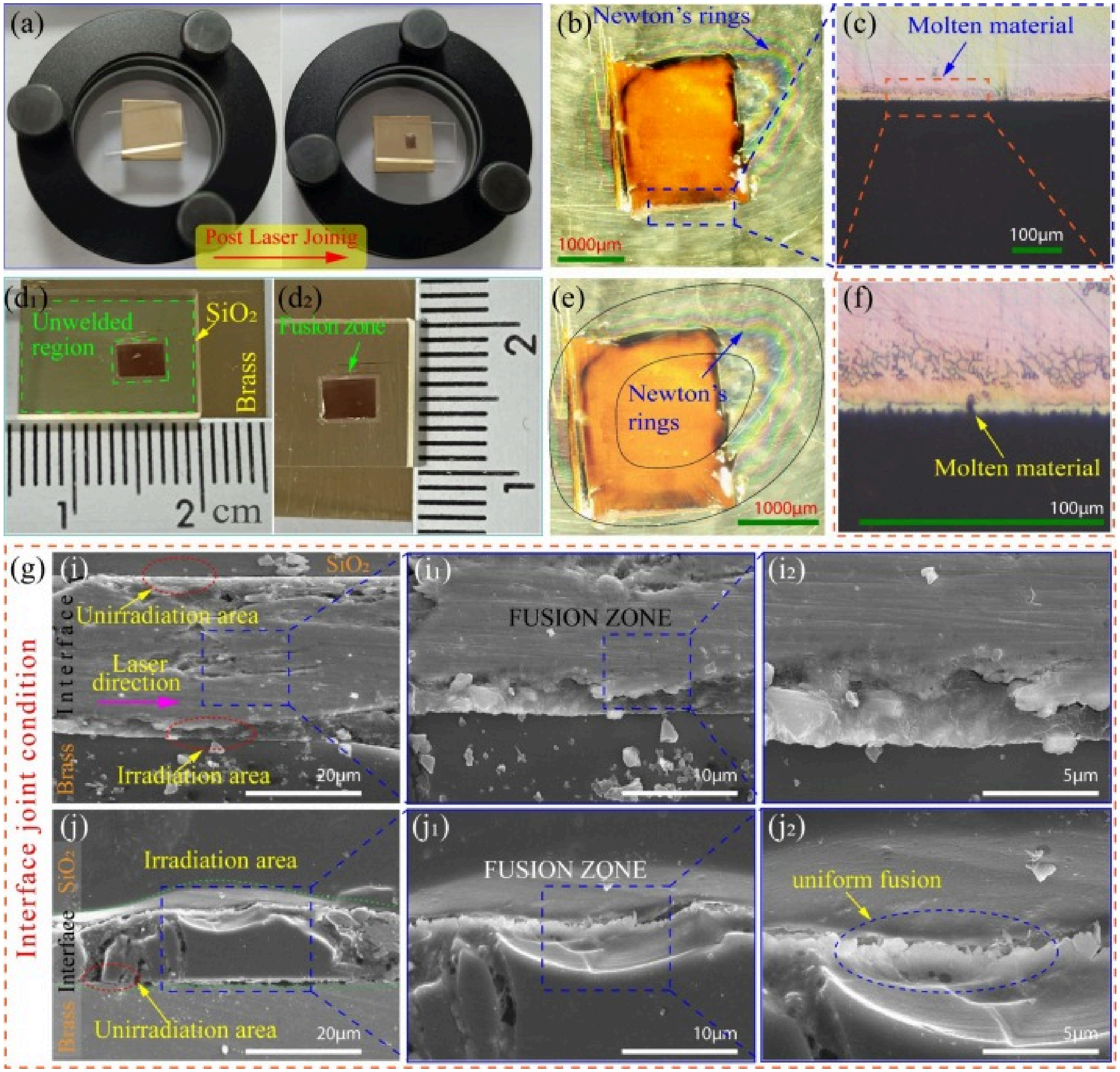

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Fig. 4. (a) Photographs of the SiO_2/brass assembly before and after laser joining, demonstrating successful bonding without external fixtures (average laser power 1.2 W, scanning speed 20 mm s^{-1}, repetition rate 150 kHz, pulse width 20–50 ns, beam diameter 3 μm, four scanning passes, line spacing 4 μm). (b) Top-view optical micrograph showing a square fusion zone surrounded by Newton's rings. (c) Optical cross-sectional image revealing localized molten and resolidified material at the SiO_2/brass interface. (d_1) Unwelded region showing the initial SiO_2/brass contact, and (d_2) welded region exhibiting a distinct fusion zone. (e) Enlarged optical image highlighting concentric Newton's rings, indicating a thin interfacial gap prior to laser irradiation. (f) High-magnification optical cross-sectional image showing molten interfacial material. (g-i) SEM cross-sectional images showing clear interfacial separation in unirradiated areas and continuous bonding within the laser-irradiated region. (j-l) High-magnification SEM images of the fusion zone revealing micro-gap collapse and a uniform, defect-free fusion morphology.

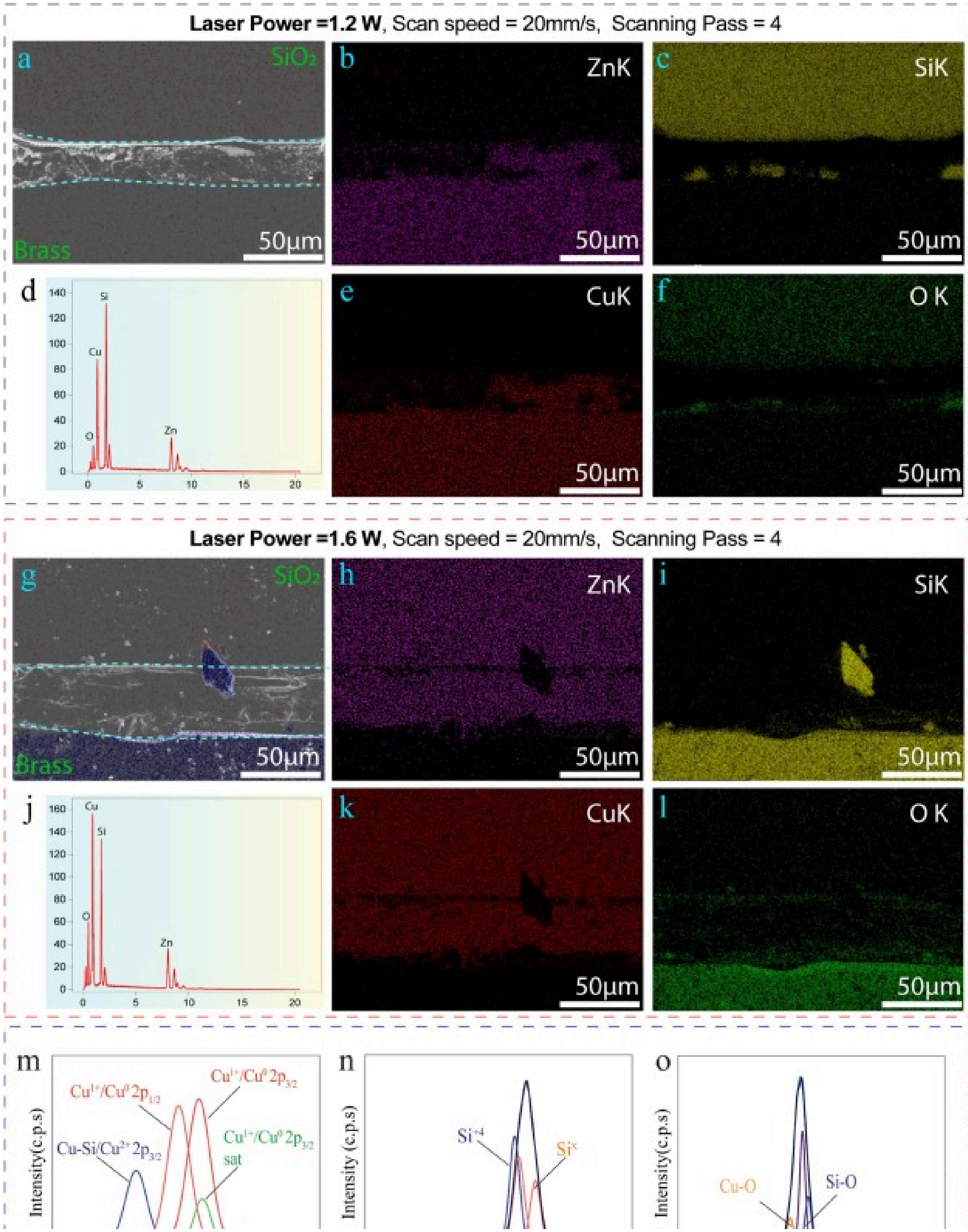
During UV laser irradiation at 355 nm wavelength, the incident laser energy is transmitted through the transparent SiO_2 substrate and primarily absorbed at the brass surface, resulting in rapid localized heating at the interface [40]. The induced thermal expansion and surface softening of the brass lead to progressive collapse of the initial micro-gap, which is evidenced by distortion and attenuation of the Newton's rings during irradiation [40]. Cross-sectional observations Fig. 4(g-i_2) and Fig. 4(g-j_2), reveal localized melting and re-solidification confined to the SiO_2/brass interface, demonstrating highly localized energy deposition without bulk damage to either material [35]. The presence of resolidified features and fine particulate structures near the interface is attributed to melt flow and partial vaporization induced by nanosecond laser pulses [11]. Previous studies have shown that vapor or plasma plume contraction during nanosecond laser irradiation can drive delayed redeposition of molten or vaporized material back onto the irradiated surface, contributing to interfacial filling [41]. Such plume-assisted redeposition mechanisms are characteristic of nanosecond laser processing under ambient conditions and can enhance joint continuity in dissimilar material welding [41]. SEM micrographs in

Fig. 4(g), confirm the formation of a continuous fusion zone in the laser-irradiated region, while a clear separation remains in the unirradiated area, indicating precise spatial control of interfacial bonding [11]. At higher magnification Fig. 4(g-j), and Fig. 4(g-j_1), the interface exhibits a smooth and uniform fusion morphology without microcracks or voids, reflecting stable thermal accumulation and controlled interfacial melting [42]. The macroscopic and microstructural observations demonstrated that UV nanosecond laser irradiation enabled effective gap collapse, localized melting, and defect-free bonding between SiO_2 and brass under optimized processing conditions. To further elucidate the interfacial bonding mechanism and the chemical interactions responsible for the observed joint integrity, the chemical composition and bonding states at the SiO_2/brass interface are examined in the following section.

4.3. Interfacial microstructure and chemical bonding of SiO_2/Brass joints

The interfacial microstructure and bonding characteristics of SiO_2/brass joints produced by UV nanosecond laser transmission welding were investigated using SEM, EDS, and XPS analyses. Due to the high optical transparency of SiO_2 at 355 nm, the incident laser radiation is transmitted through the glass and preferentially absorbed by the brass substrate, initiating localized surface heating at the metal side of the interface [11]. Heat is subsequently transferred toward the metal–glass interface via thermal conduction, leading to confined interfacial fusion.

SEM cross-sectional images shown in Fig. 5(a) and Fig. 5(g) reveal continuous fusion zones formed at the SiO_2/brass interface for both laser powers investigated (1.2 W and 1.6 W), without macroscopic cracks or interfacial voids. At 1.2 W, the fusion layer is smooth and uniform, whereas at 1.6 W localized interfacial disturbances are observed, indicating enhanced thermal accumulation at higher laser power. In both cases, the fusion region remains strictly confined to the laser-irradiated zone, confirming that joint formation is governed by localized energy deposition rather than bulk substrate heating, which is a characteristic feature of nanosecond laser transmission welding of glass–metal systems [40].



Laser Power =1.2 W, Scan speed = 20mm/s, Scanning Pass = 4
a
SiO2
Brass
50μm
b
ZnK
50μm
c
SiK
50μm
d
Si
Cu
O
Zn
e
CuK
50μm
f
O K
50μm
Laser Power =1.6 W, Scan speed = 20mm/s, Scanning Pass = 4
g
SiO2
Brass
50μm
h
ZnK
50μm
i
SiK
50μm
j
Cu
Si
O
Zn
k
CuK
50μm
l
O K
50μm
m
Intensity(c.p.s)
Cu-Si/Cu2+ 2p3/2
Cu1+/Cu0 2p1/2
Cu1+/Cu0 2p3/2
Cu1+/Cu0 2p3/2 sat
n
Intensity (c.p.s)
Si+4
Six
o
Intensity(c.p.s)
Cu-O
Si-O

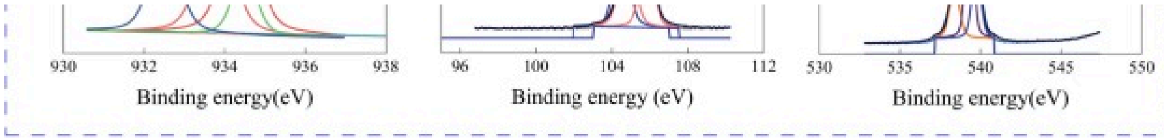


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Fig. 5. Microstructural, elemental, and chemical-state characterization of the brass/SiO_2 interface formed by UV nanosecond laser transmission welding (λ=355 nm). (a–f) SEM cross-sections, EDS elemental maps, and spectra of joints fabricated at 1.2 W, and (g–l) corresponding results at 1.6 W (v=20 mm s^{-1}, 150 kHz, four passes). Dashed lines indicate the fusion zone. Zn, Si, Cu, and O elemental maps show interfacial elemental redistribution and nanoscale intermixing. (m–o) High-resolution XPS spectra acquired from the brass-side fracture surface of the joint produced at 1.2 W: (m) Cu 2p, (n) Si 2p, and (o) O 1s spectra, confirming the formation of a chemically bonded Cu–O–Si-containing interlayer.

EDS elemental maps in Fig. 5(b-f) and Fig. 5(h-l) show pronounced elemental redistribution and nanoscale intermixing within the fusion region. Zn exhibits clear enrichment near the SiO_2/brass interface Fig. 5(b,h), while Cu remains relatively uniformly distributed within the brass substrate, preserving a continuous metallic network Fig. 5(e,k). Similar interfacial Zn enrichment has been reported in nanosecond laser joining of Zn-containing alloys and glass–metal systems [35]. The Si and O maps Fig. 5(c,i,f,l) indicate partial incorporation of Si into the fusion layer and significant oxygen enrichment at the interface, suggesting the formation of an oxide-containing interfacial region [43]. Quantitative EDS results are summarized in Table 4, confirming elevated concentrations of Cu, Zn, Si, and O within the fusion zone for both processing conditions. Stoichiometric analysis indicates that the available oxygen is insufficient to fully convert all Si into stoichiometric SiO_2, implying the formation of a chemically graded metal-oxide-silicate interlayer rather than a sharp metal/glass boundary [34].

Table 4. EDS composition of the fusion area.

Sample 1; Laser Power=1.2W, scanning speed=20mm/s, Scanning pass=4			
Element	**Weight %**	**Atomic %**	**Description**
O	13.08	32.38	Preferential oxidation of Zn and partial oxidation of Cu
Si	17.74	25.01	Partially incorporated Si; insufficient O for full SiO_2
Cu	41.59	25.91	Residual metallic brass+partial Cu_2O/CuO
Zn	27.58	16.70	Zn preferentially oxidized to ZnO; promotes wetting
Sample 2; Laser Power=1.6W, scanning speed=20mm/s, Scanning pass=4			
Element	**Weight %**	**Atomic %**	**Description**
O	15.29	34.63	Enhanced oxide formation at interface
Si	24.25	31.28	Increased Si participation in fusion layer
Cu	36.94	21.06	Continuous metallic backbone with partial oxidation
Zn	23.52	13.04	ZnO-rich interfacial wetting layer

XPS analysis was performed on the joint produced at 1.2W, which exhibited the most uniform interfacial morphology. The Cu 2p spectrum Fig. 5(m) reveals the coexistence of Cu^0/Cu^+ and Cu^{2+} states, indicating partial oxidation of copper during laser processing [40]. The Si 2p spectrum Fig. 5(n) shows a dominant Si^{4+} component together with a lower-binding-energy contribution attributed to Cu–O-Si bonding, demonstrating direct chemical interaction between oxidized copper species and the SiO_2 network [40]. The O

1s spectrum Fig. 5(o) is deconvoluted into components corresponding to CuO, Cu–O-Si, and SiO_2, providing direct chemical-state evidence for the formation of a continuous Cu-Zn-O-Si interlayer, which stabilizes the SiO_2/brass joint under nanosecond laser irradiation [43].

Furthermore, besides promoting chemical bonding, this graded Cu–Zn–O–Si interlayer acts as a transitional zone between brass and SiO_2 during cooling. Because the interfacial composition changes progressively from metallic brass to oxide/silicate species, the abrupt mismatch in thermal expansion and stiffness across the interface is partially reduced. As a result, thermally induced strain can be redistributed more gradually during rapid cooling, which helps relieve local stress concentration and suppress interfacial cracking or debonding. This stress-accommodation effect is consistent with the improved joint integrity and higher shear strength observed under optimized processing conditions [35], [43].

4.4. Parametric effect on interfacial microstructure and joint performance

Optimized laser processing parameters strongly influence interfacial microstructure and mechanical performance in SiO_2/brass joints fabricated by UV nanosecond laser transmission welding. Laser power, scanning speed, and scanning passes collectively define the transient thermal field at the interface,

thereby controlling elemental redistribution, oxide formation, and defect evolution during joining [40]. As shown in Fig. 6(a), at moderate laser power (1.2W), localized heating of the brass surface initiates selective melting and oxidation of Zn due to its lower melting point and higher chemical reactivity compared with Cu. The formation of a ZnO- and Cu_2O-rich interlayer enhances wetting at the interface with the thermally softened SiO_2 surface, resulting in the development of a dense and continuous laser-affected interfacial zone (LAIZ). This compact and well-bonded interfacial structure facilitates effective load transfer across the joint, which corresponds to the highest shear strength observed in Fig. 6(b). When the laser power is increased to 1.6–2.0W, excessive heat accumulation associated with nanosecond pulsed irradiation leads to intensified Zn evaporation and unstable molten flow behavior. These phenomena promote the formation of interfacial voids and gas entrapment. Although the apparent interfacial width increases at higher power, the presence of these defects introduces stress concentration sites, resulting in a deterioration of joint strength. Similar trends have been reported in previous studies on laser joining of metal–glass systems, where moderate energy input promoted controlled interfacial reactions and dense bonding, while excessive energy input led to elemental volatilization, defect formation, and mechanical degradation [35], [44]. These findings are consistent with the present observations, confirming that an optimal laser power is essential to balance interfacial reaction and thermal stability.

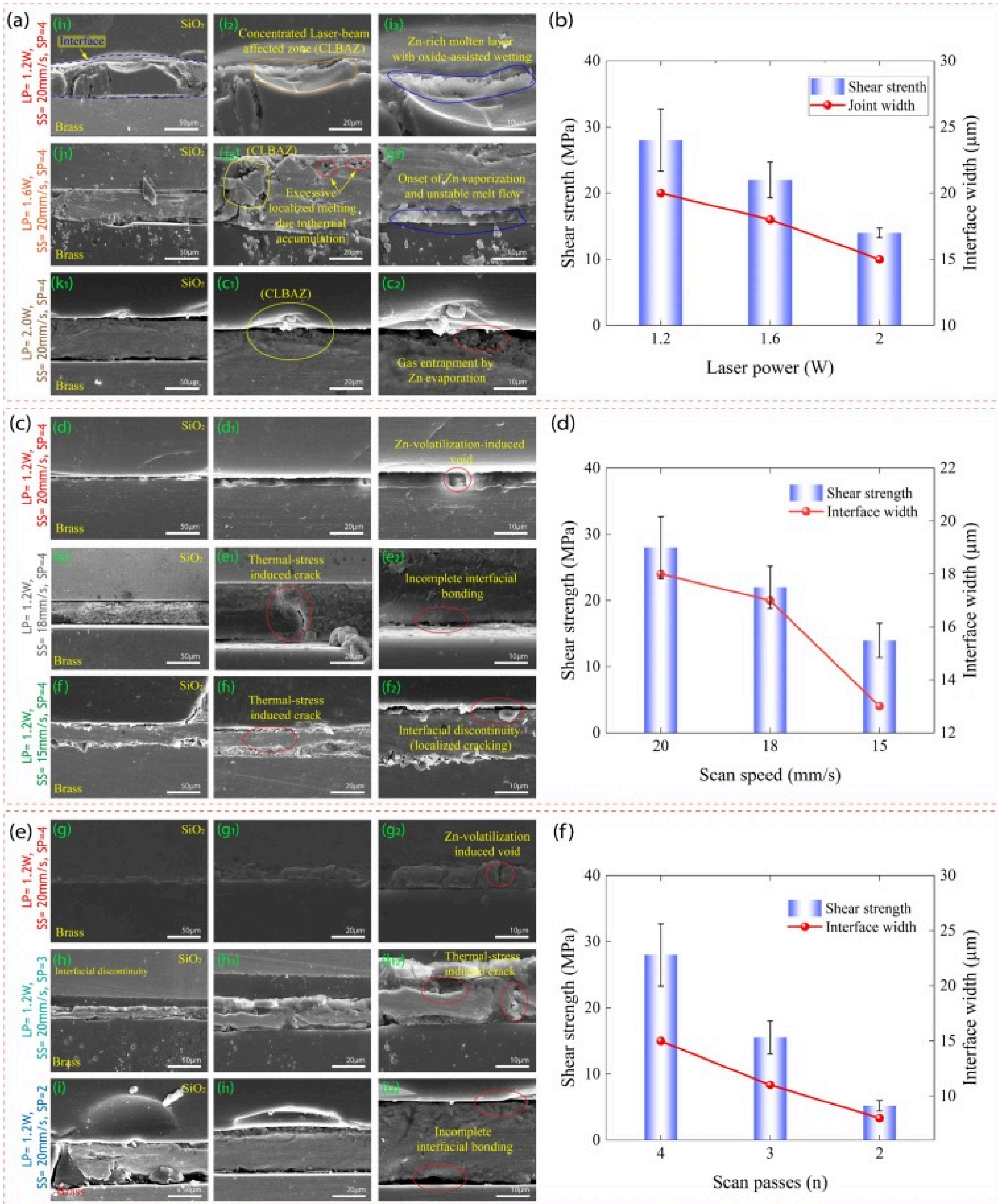

(a)
LP= 1.2W, SS= 20mm/s, SP=4
LP= 1.6W, SS= 20mm/s, SP=4
LP= 2.0W, SS= 20mm/s, SP=4
Interface
Concentrated Laser-beam affected zone (CLBAZ)
Zn-rich molten layer with oxide-assisted wetting
(CLBAZ)
Excessive localized melting due to thermal accumulation
Onset of Zn vaporization and unstable melt flow
(CLBAZ)
Gas entrapment by Zn evaporation
SiO2
Brass
(b)
Shear strenth
Joint width
Shear strenth (MPa)
Interface width (μm)
Laser power (W)
(c)
LP= 1.2W, SS= 20mm/s, SP=4
LP= 1.2W, SS= 18mm/s, SP=4
LP= 1.2W, SS= 15mm/s, SP=4
Zn-volatilization-induced void
Thermal-stress induced crack
Incomplete interfacial bonding
Thermal-stress induced crack
Interfacial discontinuity (localized cracking)
(d)
Shear strength
Interface width
Shear strength (MPa)
Interface width (μm)
Scan speed (mm/s)
(e)
LP= 1.2W, SS= 20mm/s, SP=4
LP= 1.2W, SS= 20mm/s, SP=3
LP= 1.2W, SS= 20mm/s, SP=2
Zn-volatilization induced void
Interfacial discontinuity
Thermal-stress induced crack
Incomplete interfacial bonding
(f)
Shear strength
Interface width
Shear strength (MPa)
Interface width (μm)
Scan passes (n)

Fig. 6. Microstructural evolution and mechanical performance of laser-welded SiO_2/brass joints under different processing parameters. (a) Effect of laser power (1.2, 1.6, and 2.0 W), (c) effect of scanning speed (20, 18, and 15 mm·s^{-1}), and (e) effect of scanning passes (4, 3, and 2). All cross-sectional SEM images are oriented with SiO_2 on the top and brass on the bottom. Blue dashed lines indicate the laser-affected interfacial zone (LAIZ), yellow arrows denote dominant interfacial phenomena, and red dashed circles highlight defects such as voids, cracks, and interfacial discontinuities. At moderate energy input, a continuous Zn-rich molten layer with oxide-assisted wetting forms a stable interfacial contact, whereas higher instantaneous or cumulative energy input promotes Zn volatilization, unstable melt flow, gas entrapment, and thermally induced cracking, resulting in interfacial degradation. Panels (b), (d), and (f) present the corresponding variations in shear strength and interfacial width. Bars represent the average shear strength with error bars indicating experimental scatter, while solid lines denote the interfacial width measured from cross-sectional SEM images. Interfacial width increases with increasing laser power or cumulative irradiation due to enhanced thermal accumulation, while shear strength reflects the competition between effective interfacial bonding at moderate energy input and defect-dominated failure at higher energy input.

Scanning speed regulates the interaction time between the laser beam and the material, directly influencing cumulative heat input [45]. At high scanning speed (20 mm·s^{-1}), insufficient thermal accumulation limits oxide formation and wetting, leading to incomplete bonding and reduced joint strength Fig. 6(c,d). An intermediate scanning speed (18 mm·s^{-1}) establishes a balanced thermal condition that promotes controlled oxidation and interfacial diffusion while suppressing thermal damage, producing a uniform interfacial microstructure and maximum joint strength [43]. In contrast, excessively low scanning speed (15 mm·s^{-1}) significantly increases heat accumulation, resulting in thermal stress concentration, interfacial cracking, and morphological instability [18]. Scanning passes primarily affect joint formation through repeated thermal cycling rather than instantaneous energy density [45]. An adequate number of passes enables gradual interfacial activation and the development of a thin, chemically graded Cu–Zn–O–Si interlayer, which accommodates the large thermal expansion mismatch between SiO_2 and brass [35].Insufficient passes lead to incomplete bonding due to limited interfacial reactions, whereas excessive thermal cycling promotes residual stress accumulation and crack initiation during cooling [15]. These results demonstrate that joint reliability is governed by the synergistic interaction of laser power, scanning speed, and scanning passes, highlighting the necessity of a narrow and well-controlled processing window for defect-free ceramic–metal joining [40].

The maximum joint strength obtained in this study was 28 MPa, indicating effective interfacial bonding and promising potential for glass–metal integration applications. Compared with representative reported results, this value is higher than the bond strength exceeding 10 MPa reported for nanosecond laser transmission bonding of glass to copper [11], and higher than the 20.7 MPa reported for quartz glass/aluminum joining at ultra-high scanning speed [46]. It is also slightly higher than the 25.75 MPa achieved by femtosecond laser welding of fused silica/aluminum [13], and close to the 29.4 MPa reported for picosecond laser welding of copper-coated glass [47].These comparisons indicate that the present joint strength is competitive with several reported glass–metal laser joining results, especially among nanosecond-based and related processes, although higher strengths have been achieved in some ultrashort-pulse systems and further improvement may still be possible through interlayer design and process optimization.

4.5. Fracture morphology

At a relatively low laser power of 1.2 W, the joints achieved the highest shear strength (~28 MPa). As shown in Fig. 7(a), fracture occurs predominantly along the interface, producing a smooth fracture surface characterized by a thin and discontinuous laser-induced metallurgical mixing layer (LIML) and a well-defined interfacial fracture zone (IFZ). This morphology indicates stable, pulse-controlled melting of the brass substrate and effective wetting of the thermally softened SiO_2 surface, which are essential for oxide-assisted chemical bonding in nanosecond laser joining [18], Similar confined interfacial fracture modes associated with high joint strength have been reported for copper–silica and glass–metal joints processed under moderate energy densities [48].

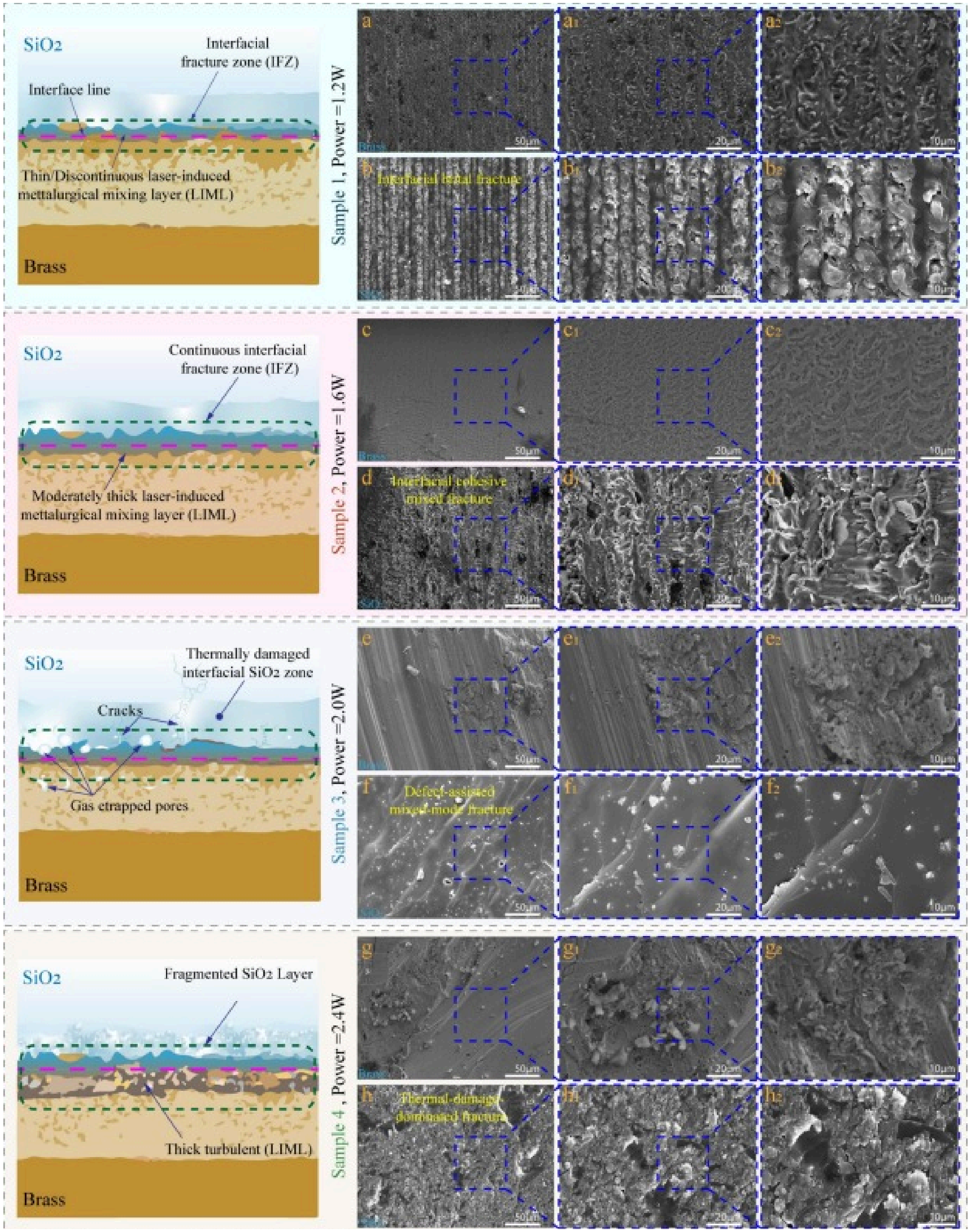

SiO2
Interfacial fracture zone (IFZ)
Interface line
Thin/Discontinuous laser-induced mettalurgical mixing layer (LIML)
Brass
Sample 1, Power =1.2W
Interfacial brittal fracture
SiO2
Continuous interfacial fracture zone (IFZ)
Moderately thick laser-induced mettalurgical mixing layer (LIML)
Brass
Sample 2, Power =1.6W
Interfacial cohesive mixed fracture
SiO2
Thermally damaged interfacial SiO2 zone
Cracks
Gas etrapped pores
Brass
Sample 3, Power =2.0W
Defect-assisted mixed-mode fracture
SiO2
Fragmented SiO2 Layer
Thick turbulent (LIML)
Brass
Sample 4 , Power =2.4W
Thermal-damage dominated fracture
50μm
20μm
10μm

Fig. 7. Schematic illustrations and corresponding SEM fracture morphologies of SiO_2–brass joints fabricated at different laser powers, while maintaining a constant scanning speed of 20 mm s^{-1} and 4 scanning passes. (a) 1.2 W: Predominantly interfacial fracture with a smooth morphology, characterized by a thin and discontinuous laser-induced metallurgical mixing layer (LIML) and a well-defined interfacial fracture zone (IFZ), corresponding to stable pulse-controlled melting and minimal thermal damage. (b) 1.6 W: Interfacial–cohesive mixed fracture with a continuous IFZ and moderately thick LIML, indicating enhanced thermal accumulation accompanied by increased melt flow and localized stress concentration. (c) 2.0 W: Thermally damaged interfacial SiO_2 zone with gas-entrapped porosity and microcracks, leading to defect-assisted fracture. (d) 2.4 W: Thermal-damage-dominated fracture characterized by fragmented SiO_2 and a thick, turbulent LIML, resulting from excessive energy input and unstable melting behavior.

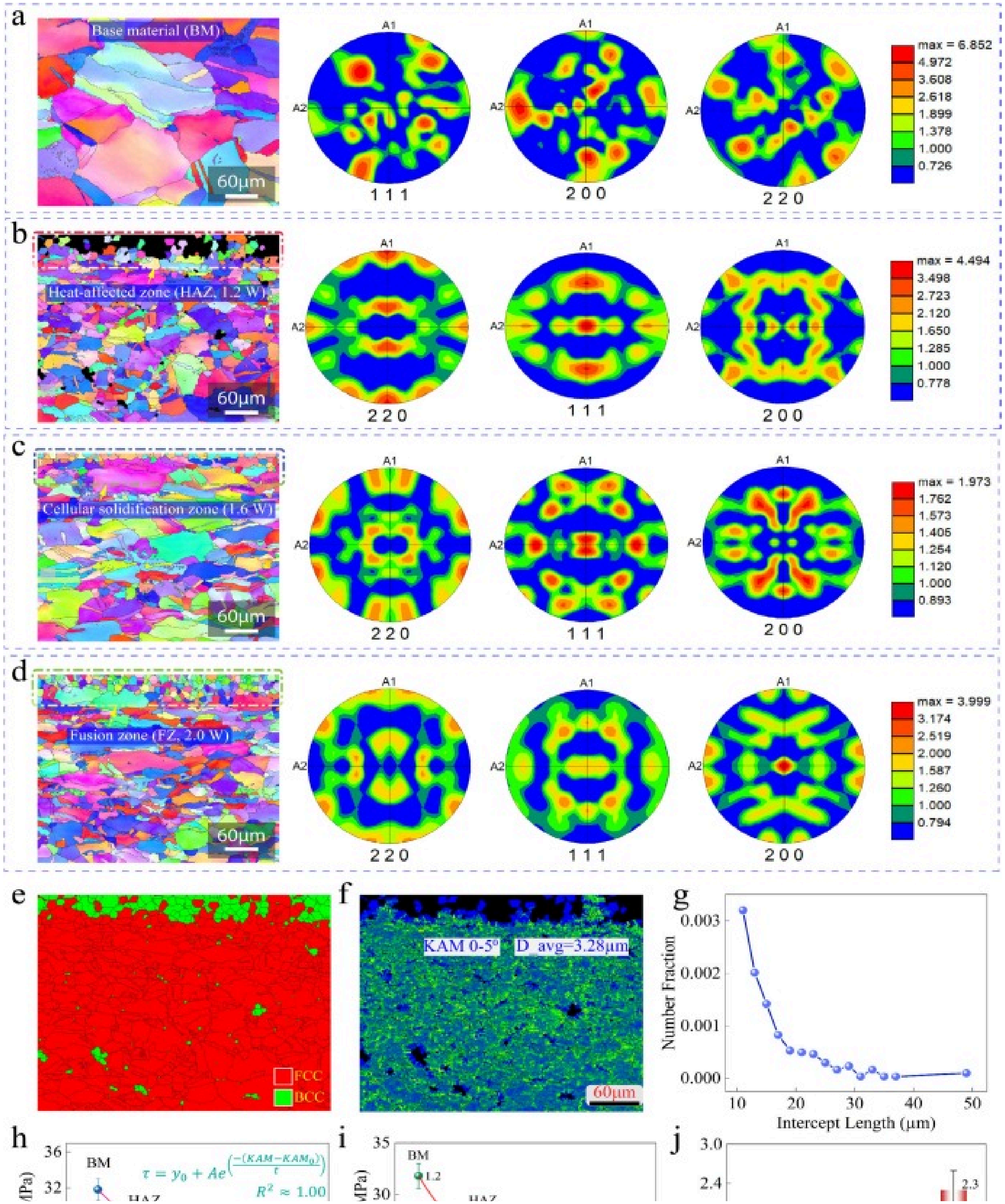
When the laser power was increased to 1.6 W, enhanced thermal accumulation intensified molten metal flow and interfacial reactions. As illustrated in Fig. 7(b), the fracture surface becomes rougher and exhibits a continuous IFZ with a moderately thick LIML. Localized melt pits aligned along the scanning direction are observed, reflecting dynamic molten pool behavior induced by cumulative pulse heating [40]. This fracture morphology corresponds to an interfacial–cohesive mixed fracture mode, where increased interfacial bonding is partially offset by non-uniform re-solidification and localized stress concentration, resulting in a reduced shear strength of ~22 MPa. Similar transitions from interfacial to mixed fracture with increasing laser energy have been widely reported in nanosecond laser joining studies [49].

At higher laser powers (≥2.0 W), joint performance degraded sharply, with shear strength decreasing to ~14 MPa at 2.0 W. As shown in Fig. 7(c) and Fig. 7(d), fracture surfaces exhibit extensive gas-entrapped porosity, microvoids, interfacial microcracks, and fragmented SiO_2 regions. These features are indicative of unstable melting, Zn volatilization, and rapid re-solidification under excessive thermal accumulation [15]. Crack initiation at pores and microcracks promotes a defect-assisted, thermal-damage-dominated fracture mode, which overwhelms the benefits of enhanced interfacial reactions and leads to premature failure. Similar defect-controlled fracture mechanisms at high energy densities have been reported in laser welding of glass–metal systems [50]. The evolution from interfacial to defect-dominated fracture demonstrates that joint strength is maximized only within a narrow laser energy window, beyond which excessive thermal accumulation governs failure. To clarify the microstructural origins of this behavior, the crystallographic characteristics of the interfacial brass region are analyzed in the following section using EBSD. The fracture behavior of the SiO_2/brass joints exhibits a strong dependence on laser energy input, confirming that joint integrity is governed by a well-defined processing window in UV nanosecond laser transmission welding. Within this window, interfacial reactions are sufficiently activated to promote chemical bonding and mechanical interlocking, while excessive thermal damage and defect formation are suppressed, as commonly observed in glass–metal laser joining systems [43].

4.6. Grain structure characterization

The crystallographic evolution of the brass adjacent to the welded interface is directly governed by the laser-induced thermal field generated during UV nanosecond laser irradiation, which imposes strong thermal gradients and non-equilibrium solidification conditions [51].

Fig. 8 summarizes the EBSD-based correlation between laser-energy-dependent microstructural evolution and shear strength degradation in laser jointed SiO_2/brass. As shown in Fig. 8(a), the base brass material (BM) exhibits coarse equiaxed grains with weak, near-random crystallographic texture and low kernel average misorientation (KAM), indicating a thermally relaxed FCC microstructure with minimal stored lattice strain [51]. Following laser irradiation at low power (1.2W), the heat-affected zone shown in Fig. 8(b) displays moderate grain refinement while maintaining low KAM values, reflecting controlled thermal exposure and limited accumulation of geometrically necessary dislocations (GNDs) [52]. This microstructural condition corresponds to the highest measured shear strength Fig. 8(h), demonstrating that moderate thermal input effectively activates interfacial bonding without inducing significant residual damage [53]. With increasing laser power to 1.6W, a cellular solidification regime develops Fig. 8(c), characterized by elongated grains aligned with the heat-flow direction and pronounced texture sharpening in the {111}, {200}, and {220} pole figures, which is characteristic of directional solidification under elevated thermal gradients in FCC alloys [54], Concurrently, the increase in KAM values Fig. 8(j), indicates enhanced lattice distortion and residual strain accumulation caused by constrained solidification and thermo-mechanical mismatch between brass [55]. This strain accumulation marks the onset of damage-controlled mechanical behavior, as reflected by the monotonic reduction in shear strength with increasing KAM Fig. 8(h) [54]. At higher laser power (2.0W), the fusion zone shown in Fig. 8(d) exhibits dendritic solidification, strong crystallographic texture, and the highest KAM levels, signifying severe thermal gradients, high GND density, and elevated residual stress [56]. Correspondingly, the normalized damage index (NDI) derived from void area fraction increases sharply under these Fig. 8(i), confirming that joint strength degradation is governed by the combined effects of strain-controlled damage and defect-controlled mechanisms such as porosity and micro-cracking [57]. The evolution of a transition from strain-tolerant equiaxed microstructures to damage-dominated dendritic regimes as laser energy input increases [58].


a
Base material (BM)
60μm
A1
A2
1 1 1
2 0 0
2 2 0
max = 6.852
4.972
3.608
2.618
1.899
1.378
1.000
0.726
b
Heat-affected zone (HAZ, 1.2 W)
max = 4.494
3.498
2.723
2.120
1.650
1.285
1.000
0.778
c
Cellular solidification zone (1.6 W)
max = 1.973
1.762
1.573
1.405
1.254
1.120
1.000
0.893
d
Fusion zone (FZ, 2.0 W)
max = 3.999
3.174
2.519
2.000
1.587
1.260
1.000
0.794
e
FCC
BCC
f
KAM 0-5°
D_avg=3.28μm
g
Number Fraction
Intercept Length (μm)
h
BM
HAZ
$\tau = y_0 + Ae^{\left(\frac{-(KAM-KAM_0)}{t}\right)}$
$R^2 \approx 1.00$
i
BM
1.2
HAZ
j
2.3

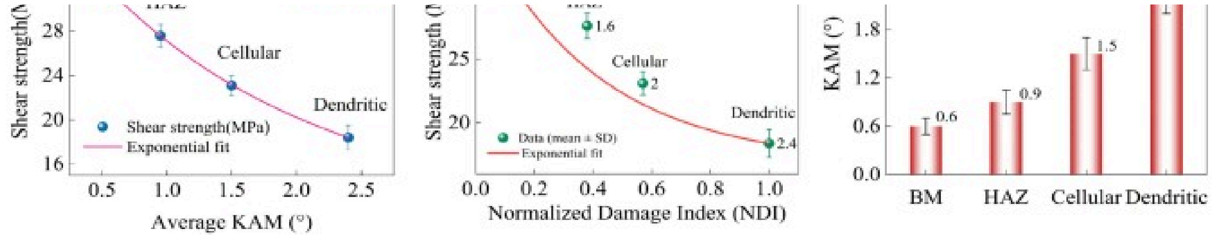


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Fig. 8. Evolution of brass microstructure, crystallographic texture, and damage accumulation across the laser-joined and mechanical response (a) EBSD inverse pole figure (IPF) map and corresponding {111}, {200}, and {220} pole figures of the base material (BM).(b) IPF map and pole figures of the heat-affected zone (HAZ) produced at a laser power of 1.2W. (c) IPF map and pole figures of the cellular solidification zone formed at a laser power of 1.6W. (d) IPF map and pole figures of the fusion zone (FZ) obtained at a laser power of 2.0W. (e)EBSD phase map showing the distribution of FCC and BCC phases. (f) Kernel average misorientation (KAM) map (0–5°) with the corresponding average grain diameter. (g)Intercept length distribution derived from EBSD data. (h)Relationship between joint shear strength and average KAM. (i) Relationship between joint shear strength and normalized damage index (NDI). (j) Average KAM values for BM, HAZ, cellular, and dendritic regions.

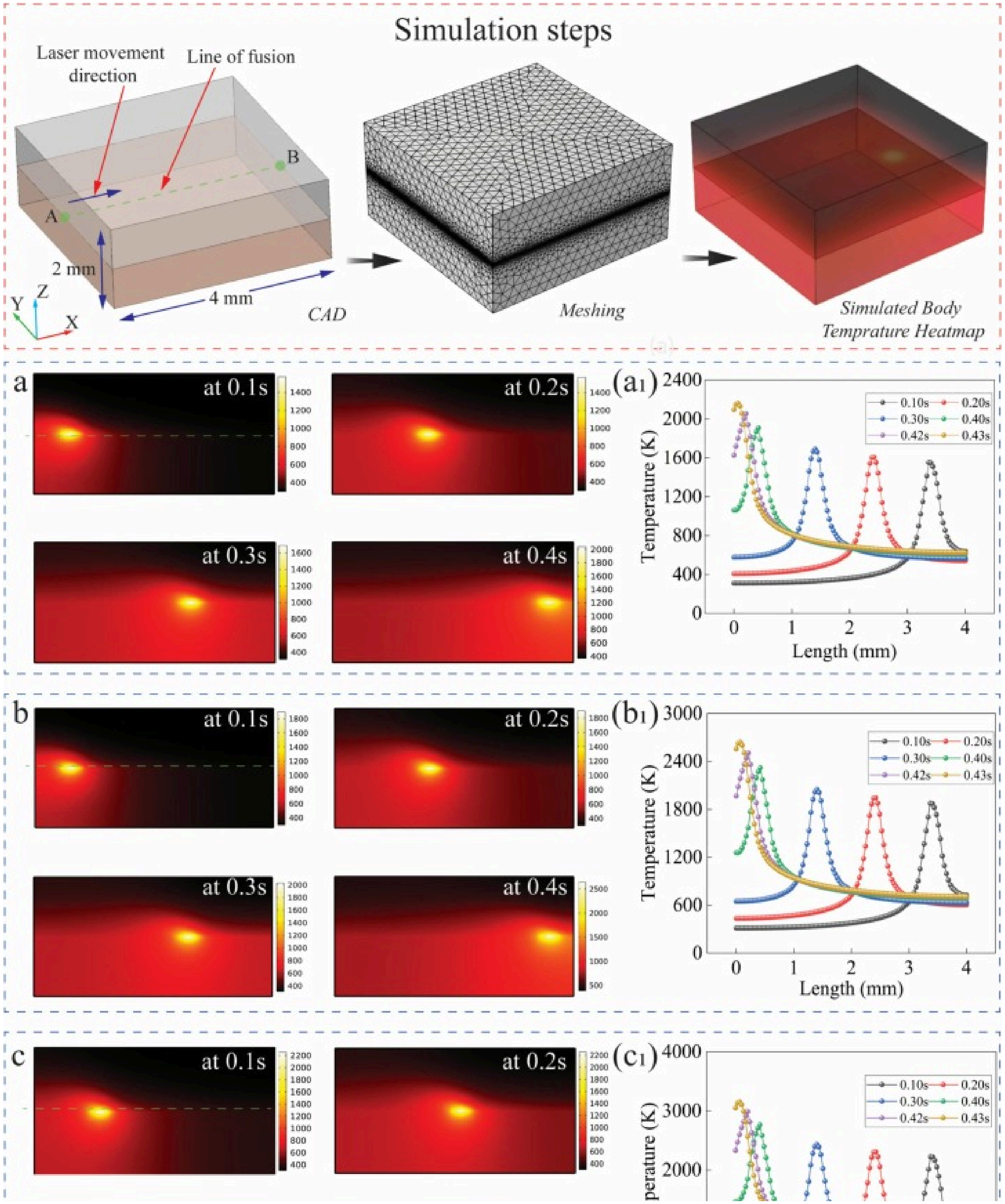
The kernel average misorientation (KAM) map shown in Fig. 8(f) exhibits predominantly low misorientation values (KAM<5°), indicating limited residual plastic strain and effective strain accommodation during cooling. The grain size distribution obtained from EBSD linear intercept analysis is presented in Fig. 8(g), which plots the number fraction of grain intercepts (y-axis) as a function of intercept length in micrometers (x-axis). The distribution is concentrated in the short intercept range (approximately 10–15 μm), with progressively decreasing frequency at larger intercept lengths. This right-skewed profile indicates a fine-grained microstructure without abnormal grain coarsening. The calculated average grain diameter is approximately 3.28 μm, confirming significant grain refinement in the fusion zone under the optimized laser condition, confirming significant grain refinement in the fusion zone. The combined presence of fine grains, high boundaries HAGB and controlled residual strain provides a microstructural basis for the enhanced mechanical stability and improve crack resistance of the optimized joint [59]. Phase mapping in Fig. 8(e) confirms that the welded brass remains predominantly face-centered cubic (FCC), with only limited and localized body-centered cubic (BCC) regions formed as a result of thermal cycling and non-equilibrium solidification during UV nanosecond laser processing. Such phase stability is characteristic of Cu-based alloys subjected to rapid heating and cooling and indicates that the optimized processing window effectively suppresses extensive phase transformation [60].

4.7. Heat accumulation and thermal saturation during joining

During UV nanosecond (355 nm) laser transmission welding of SiO_2/brass joints, interfacial temperature evolution is dominated by cumulative heat accumulation rather than single-pulse heating, owing to the high repetition rate and strong pulse overlap inherent to nanosecond irradiation [50]. To quantify this behavior, transient finite-element simulations were performed using COMSOL Multiphysics, employing a moving Gaussian surface heat source, which has been widely adopted for modeling conduction-mode nanosecond laser processing of metals and dissimilar material systems [40].

Direct experimental validation of the transient temperature field was not performed in this study. The 355 nm nanosecond laser induces highly localized and rapid thermal cycles, and at a repetition rate of 150 kHz the inter-pulse interval is only about 6.7 μs, making reliable real-time temperature measurement highly challenging. Moreover, the optical arrangement and jig system required for Newton-ring-based interfacial alignment prevented reliable in-situ IR observation of the joining interface. Therefore, the simulation results are used primarily to reveal the relative thermal evolution and comparative trends under different processing conditions, rather than to provide exact absolute temperature values.

At a repetition rate of 150 kHz and a scanning speed of 20 mm s^{-1}, the pulse overlap exceeds 95%, significantly suppressing thermal relaxation between successive pulses. Under such conditions, the interfacial temperature rise is primarily governed by average laser power rather than individual pulse energy, consistent with thermal accumulation regimes reported for nanosecond laser welding of glass-metal systems [50]. The simulated transient temperature fields along a 4 mm scan line at laser powers of 1.2, 1.6, and 2.0 W are shown in Fig. 9(a) and Fig. 9(c), with the corresponding temperature profiles extracted along the scanning direction presented in Fig. 9(a_1) and Fig. 9(c_1).


Simulation steps
Laser movement direction
Line of fusion
A
B
2 mm
4 mm
CAD
Meshing
Simulated Body Temprature Heatmap
a
at 0.1s
at 0.2s
at 0.3s
at 0.4s
(a1)
Temperature (K)
Length (mm)
0.10s
0.20s
0.30s
0.40s
0.42s
0.43s
b
(b1)
c
(c1)

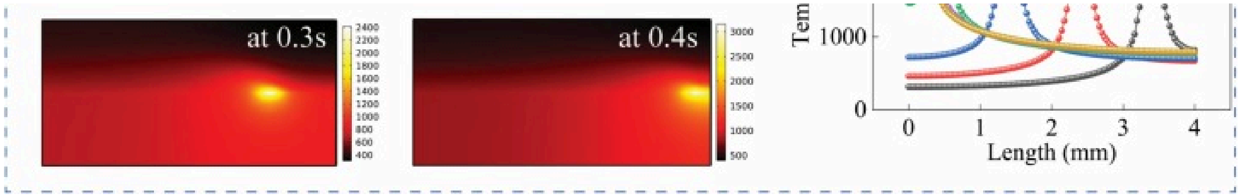


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Fig. 9. Finite-element simulation of transient thermal behavior during laser welding. (Top row) Schematic illustration of the simulation procedure, including the CAD geometry and laser scanning configuration, finite-element meshing, and the resulting simulated 3D temperature field induced by a moving Gaussian heat .
Source. The laser scans along a 4 mm line at the SiO_2-brass interface, as indicated by the arrow. (a-c) Transient temperature distributions along a representative single laser scan line at laser powers of (a) 1.2 W, (b) 1.6 W, and (c) 2.0 W, respectively. For each laser power, temperature fields are shown at t = 0.1 s, 0.2 s, 0.3 s, and 0.4 s,42s and 43s corresponding to a total scan time of 0.44 s with an effective scanning speed of 10 mm s^{-1}. The dashed line denotes the SiO_2-brass interface, and the arrow indicates the laser scanning direction. (a1–c1) Corresponding temperature profiles extracted along the laser scanning direction at different times for laser powers of 1.2 W, 1.6 W, and 2.0 W, respectively, illustrating the temporal evolution of peak temperature and heat accumulation during laser scanning. The simulations reveal progressively increased peak temperatures and extended high-temperature regions with increasing laser power, reaching approximately 2200 K, 2600 K, and 3200 K for 1.2 W, 1.6 W, and 2.0 W, respectively

At 1.2W Fig. 9(a) and Fig. 9(a_1), a localized high-temperature zone forms near the laser–material interaction region, reaching peak temperatures on the order of $\sim 2.2\times10^3$ K while remaining spatially confined along the scanning path. Such spatial confinement of the thermal field is characteristic of conduction-dominated nanosecond laser processing and indicates controlled localized heating with limited lateral heat diffusion [11]. This thermal regime favors effective interfacial activation and stable localized melting of the brass substrate, which has been reported to promote continuous reaction-layer formation and high joint strength in nanosecond laser glass–metal joining systems [61].

Increasing the laser power to 1.6W Fig. 9(b) and Fig. 9(b_1), results in higher peak temperatures (approaching ~2.6×10^3 K) and a pronounced extension of the high-temperature region along the scanning direction, reflecting enhanced thermal accumulation and steeper temperature gradients. Similar expansion of the thermal field with increasing average laser power has been reported in nanosecond laser welding and is associated with intensified interfacial diffusion and metallurgical interaction [40]. However, the broader thermal field indicates that the process approaches the upper limit of stable joining, consistent with previous observations that excessive thermal accumulation reduces joint strength due to non-uniform melting and stress concentration [62].

At 2.0W Fig. 9(c) and Fig. 9(c_1), further increases in laser power primarily expand the heated volume while the peak temperature increases only marginally, indicating the onset of conduction-controlled thermal saturation. Such saturation behavior, in which additional energy is dissipated into the surrounding substrate rather than increasing peak temperature, has been widely reported in high-repetition-rate nanosecond laser processing [63]. This thermal saturation is commonly associated with unstable melting behavior and joint degradation at excessive energy input [64].

The temporal temperature histories exhibit a rapid initial rise followed by a quasi-steady plateau, confirming the transition from transient heating to accumulation-dominated thermal behavior. The simulations reveal a narrow thermal processing window in which sufficient heat accumulation enables effective interfacial activation and localized brass melting, whereas excessive energy input leads to thermal saturation and deteriorated joint integrity.

After the laser beam passes the sampling location, the joint undergoes a rapid transition from localized heating to conduction-dominated cooling, which is a typical thermal response in laser-based joining processes [65]. As shown in Fig. 10, points closer to the SiO_2/brass interface exhibit higher peak temperatures and faster initial cooling rates due to steeper local temperature gradients generated by nanosecond-scale energy deposition [62]. With increasing cooling time, thermal diffusion into the surrounding glass and brass substrates progressively reduces these gradients, leading to convergence of the temperature profiles at different depths, as widely reported in laser welding and laser transmission joining studies [66]. This behavior reflects a diffusion-controlled cooling regime governed by temperature gradients and thermal conductivity, consistent with Fourier heat conduction behavior, and is beneficial for mitigating residual thermal stresses and enhancing the structural integrity of glass-metal joints [40].

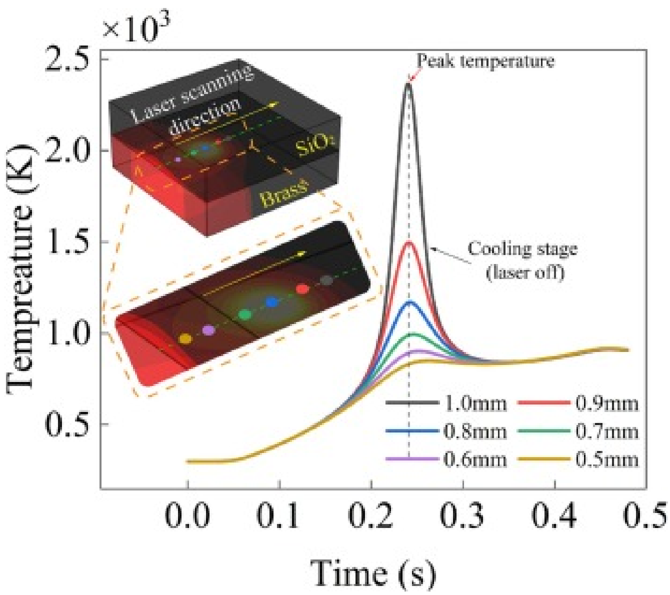

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Fig. 10. Cooling-stage thermal history of representative points along the thickness direction during UV nanosecond laser transmission welding of SiO_2/brass. Sampling points (1.0–0.5 mm from the glass–metal interface) are indicated in the inset. The dashed

line marks the peak temperature, after which conduction-dominated cooling leads to gradual convergence of the temperature profiles at different depths.

5. Validation of model

5.1. Validation of the thermal model

The validity of the proposed bulk-averaged thermal model was assessed by comparing the predicted temperature distributions with experimentally observed interfacial morphology and independently simulated transient thermal behavior. Fig. 11(a–d) summarizes the calculated temperature fields obtained directly from the MATLAB implementation of Eqs. (6)-(12), including temperature variation across the interface thickness, along the scan length, and in three-dimensional space. Fig. 11(a) shows the calculated temperature profiles normal to the SiO_2/brass interface for laser powers of 1.2, 1.6, and 2.0W. The model predicts a pronounced asymmetry in thermal penetration, characterized by a steep temperature decay in the glass and a more gradual decay into the brass substrate. This behavior directly reflects the asymmetric thermal diffusivities of SiO_2 and brass and is consistent with the experimentally observed confinement of melting to a narrow interfacial region. The predicted peak bulk-averaged interfacial temperatures increase monotonically with laser power, reaching approximately $\mathbf{2.2 \times 10^3}$ K at 1.2W, $\mathbf{2.6 \times 10^3}$ K at 1.6W, and exceeding $\mathbf{3.0 \times 10^3}$ K at 2.0W. These values are sufficient to induce localized melting of the brass surface while remaining compatible with the absence of bulk thermal damage in the SiO_2 substrate under optimized conditions.

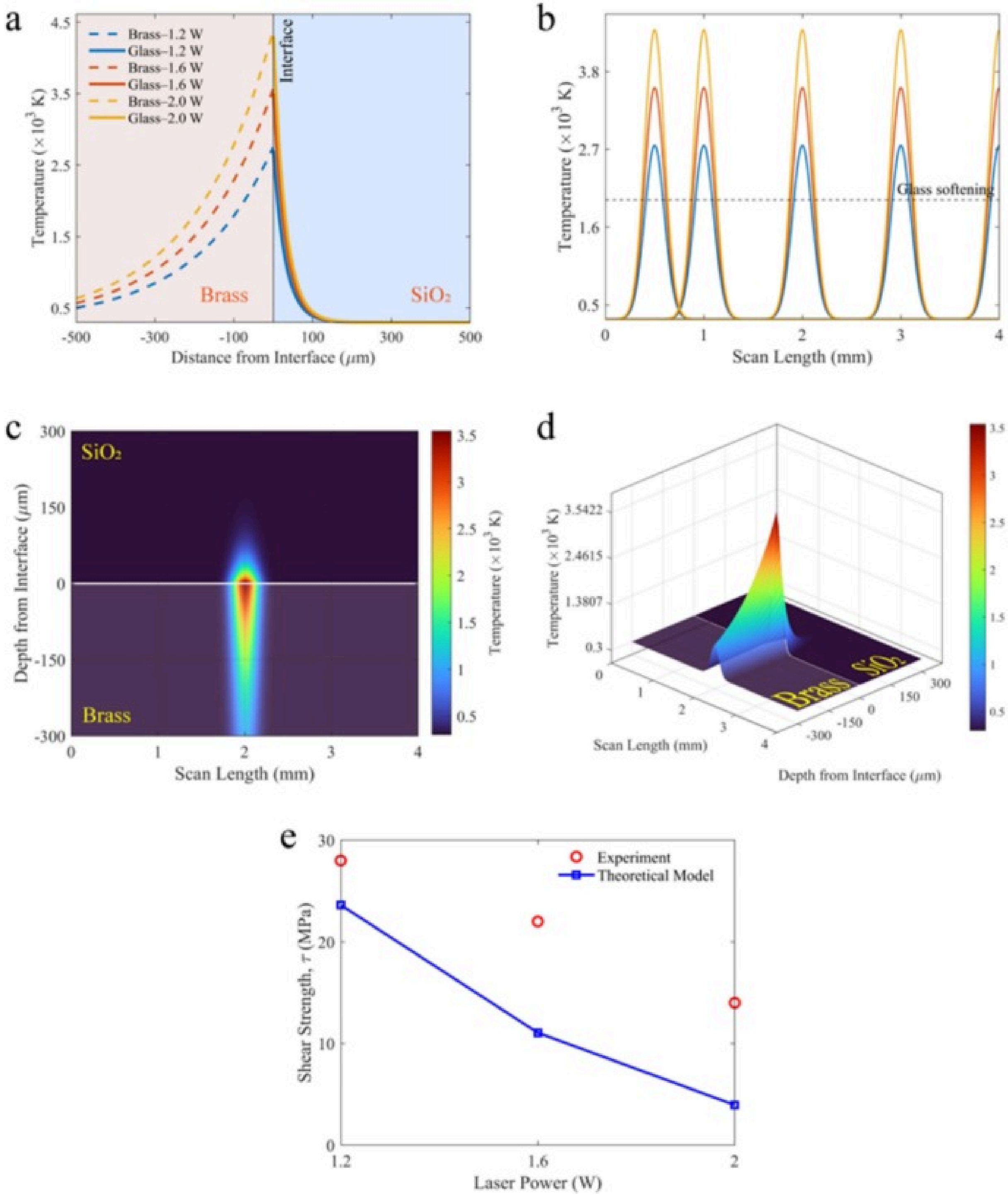
a
Brass–1.2 W
Glass–1.2 W
Brass–1.6 W
Glass–1.6 W
Brass–2.0 W
Glass–2.0 W
Interface
Brass
SiO₂
Temperature (×10³ K)
Distance from Interface (μm)
b
Glass softening
Temperature (×10³ K)
Scan Length (mm)
c
SiO₂
Brass
Depth from Interface (μm)
Scan Length (mm)
Temperature (×10³ K)
d
Temperature (×10³ K)
3.5422
2.4615
1.3807
0.3
Brass
SiO₂
Scan Length (mm)
Depth from Interface (μm)
e
Experiment
Theoretical Model
Shear Strength, τ (MPa)
Laser Power (W)

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Fig. 11. Validation of the thermal–mechanical model for UV nanosecond laser welding of SiO_2/brass interfaces. (a) Bulk-averaged temperature profiles normal to the SiO_2/brass interface for laser powers of 1.2, 1.6, and 2.0W, showing asymmetric thermal penetration into brass and rapid decay in the glass. (b) Predicted temperature evolution along the scan direction at multiple time snapshots, illustrating pulse-overlap-induced thermal accumulation; the dashed line indicates the glass softening temperature. (c) Two-dimensional interfacial temperature field at 1.6W, showing localized heating along the scan path. (d) Corresponding three-dimensional bulk-averaged temperature field, highlighting confined interfacial heating and asymmetric diffusion. (e) Comparison of experimental and predicted shear strength as a function of laser power, demonstrating non-monotonic strength evolution governed by thermal accumulation and damage.

The evolution of temperature along the scan direction is shown in Fig. 11(b). The model reproduces a series of spatially periodic temperature peaks corresponding to successive laser passes, with incomplete cooling between peaks. This behavior confirms that interfacial heating is dominated by pulse-overlap-induced thermal accumulation rather than isolated single-pulse heating. As laser power increases, the temperature peaks broaden and the baseline temperature rises, indicating progressive accumulation and the approach toward thermal saturation. This trend is consistent with the experimentally observed transition from stable bonding to thermally induced defects at excessive energy input. Fig. 11(c) and Fig. 11(d) present the two-dimensional and three-dimensional temperature fields predicted by the model for a representative laser power of 1.6W. The calculated temperature field exhibits a narrow, spindle-shaped high-temperature region localized at the interface and aligned with the scan path. This spatial confinement agrees well with cross-sectional optical and SEM observations of the fusion zone and supports the use of a bulk-averaged joining temperature as the governing thermal parameter. Importantly, the model captures the saturation-like behavior at high power, where further increases in laser power primarily expand the heated volume rather than significantly increasing the peak temperature, consistent with conduction-controlled heat dissipation.

5.2. Validation of the shear-strength model

The shear-strength model was validated by direct comparison between experimentally measured joint strength and model predictions derived from the thermal history, as shown in Fig. 11(e). Experimentally, the shear strength of SiO_2/brass joints exhibits a clear non-monotonic dependence on laser power, with a maximum value of approximately 28 MPa at 1.2 W, followed by a reduction to about 22 MPa at 1.6 W and a further decrease to approximately 14 MPa at 2.0 W.

The proposed model successfully reproduces both the magnitude and trend of the experimental data. At low laser power (1.2 W), the model predicts high shear strength due to sufficient glass softening, extended time above the softening temperature, and an effective bonding width that promotes strong interfacial bonding while limiting residual damage. At intermediate power (1.6 W), the predicted strength decreases as residual thermal stress and strain accumulation begin to offset the benefits of enhanced bonding, consistent with the onset of cellular solidification and increased lattice distortion observed by EBSD. At high power (2.0 W), the model predicts pronounced strength degradation as damage-related terms dominate, reflecting the experimentally observed porosity formation, microcracking, and damage-assisted fracture. The agreement between predicted and measured shear strength confirms that the shear-strength model correctly captures the competition between bonding enhancement and damage accumulation. Notably, the thermal expansion parameter is invoked only in the residual-stress-based damage term and does not influence the thermal field, ensuring a clear separation between thermal and mechanical effects. This coupling strategy enables accurate prediction of strength degradation without artificially modifying the temperature distribution.

6. Conclusions

Based on the combined experimental and numerical investigation, the main conclusions of this study are summarized as follows:

I. This study clarified the thermal, microstructural, and mechanical mechanisms governing UV nanosecond laser transmission welding of SiO_2/brass interfaces through a combined experimental–numerical approach. Thermal simulation showed that, at a repetition rate of 150 kHz, interfacial heating is governed mainly by average laser power rather than individual pulse energy, with the peak interfacial temperature increasing from about 2200 K at 1.2 W to about 3200 K at 2.0 W.

II. Further increase in laser energy mainly expanded the heated volume rather than proportionally increasing the peak temperature, indicating conduction-controlled thermal saturation and the existence of a narrow processing window. Within this window, localized melting of the brass surface promoted collapse of the initial nanometric interfacial gap and formation of a continuous fusion zone, enabling effective joining without excessive thermal damage to the SiO_2 substrate.

III. Chemical and elemental analyses confirmed the formation of a thin Cu-Zn-O-Si interlayer, where preferential oxidation of Zn improved wetting and facilitated Cu–O-Si chemical bonding, indicating that the joint was formed through metallurgical bonding rather than simple mechanical interlocking. EBSD analysis further revealed that controlled thermal accumulation produced a refined and strain-tolerant brass microstructure near the interface, characterized by fine grains (about 3.3 μm), low KAM values, and a high fraction of high-angle grain boundaries, which helped accommodate thermal strain during cooling and suppress crack initiation.

IV. Excessive thermal input caused a transition to cellular and dendritic solidification, accompanied by increased KAM, a higher normalized damage index, and porosity formation, all of which degraded joint integrity. As a result, the joint shear strength showed a clear non-monotonic dependence on laser power, reaching a maximum of about 28 MPa at 1.2 W and then decreasing significantly at higher powers because of damage-dominated fracture. Overall, the results demonstrate that controlled thermal accumulation, rather than maximum energy input, is the key factor for achieving high-strength and reliable glass-metal joints, and the proposed thermal–mechanical framework provides useful guidance for process optimization in microelectronic and optoelectronic applications.

CRediT authorship contribution statement

Mohammed Omar Abdullah Ba Matraf: Writing – original draft, Visualization, Investigation, Data curation, Conceptualization. **Yongling Wu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Hongyu Zheng:** Writing – review & editing, Supervision, Resources.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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